

CANCER RESEARCH CENTRE

KOCHI

THESIS REPORT

Submitted in partial fulfilment of the requirement for the award of Degree

BACHELOR of ARCHITECTURE



VINAY DILIP J

USN: 1BQ14AT075

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B M S SCHOOL OF ARCHITECTURE

Yelahanka, Bangalore

(Affiliated to Visvesvaraya Technological University, Belgaum)



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ARCHITECTURAL DESIGN PROJECT
CANCER RESEARCH CENTRE- KOCHI

VINAY DILIP J

1BQ14AT075, 8 SEM



BMS SCHOOL OF ARCHITECTURE, Bangalore
(Affiliated to Visvesvaraya Technological University, Belgaum)

Certificate

This is to certify that this thesis report titled, “*CANCER RESEARCH CENTRE - KOCHI*”, is a bonafide work carried out by **VINAY DILIP J**, bearing USN **1BQ14AT075**, in partial fulfilment of the requirements for award of the under-graduate degree of **Bachelor of Architecture**.

Thesis Guide

Asst. Prof. Mondira Ganguly

Thesis Co-ordinator

Assoc. Prof. Syed M Sarmad

Director, BMSSA

Prof. Dr. Sapna Papu

Examined by (name, signature and date)

1. Ext

2. Ext

3. Int

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Declaration

This thesis titled “*CANCER RESEARCH CENTRE - KOCHI*”, submitted in partial fulfilment of the requirement for award of the under-graduate degree of Bachelor of Architecture is my original work to the best of my knowledge.

The sources for the various information and data used have been duly acknowledged.

The work has not been submitted or provided to any other institution/organization for any diploma/degree or any other purpose.

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VINAY DILIP J

USN: 1BQ14AT075

BMS School Of Architecture, Bangalore

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VINAY DILIP J

USN: 1BQ14AT075

BMS School of Architecture, Bangalore

(Affiliated to Visvesvaraya Technological University, Belgaum)

ABSTRACT

In an era where more time is spent indoors, making the interior space healthy and comfortable becomes critical, especially with regards to healthcare facilities. It is not only the physical benefits, well designed healthcare space will improve the psychological health of patients. An understanding of therapeutic architecture is relevant in analysing the impact of various elements such as lighting, colours, landscaping etc. on health and how it relates to the design of healthcare centres, specifically that of cancer treatment. Cancer is a deadly disease that effects millions of people around the world. The project aims at designing a healthcare environment that help patients feel comfortable, both mentally and physically. Primary case-based research on Adayar Cancer Institute, Chennai and Madras medical college, Chennai and Vedanta Cancer Centre, Raipur and secondary case study on Carti Cancer Centre, Little rock, Windsor regional cancer centre, Ontario and Centre for Cancer And Health, Copenhagen is critically investigated.

The field of allopathic medicine, the health care system and delivery mechanism in India are pluralistic in nature, with many different kinds of healthcare providers (public and private) in different settings, with different delivery schemes providing cure and treatment for the patient's illness. Cancer is one among the three deadly non-communicable diseases classified in India by the Ministry of Health and Family Welfare with an estimated 53% of annual deaths and is still considered as a life-threatening disease (ICMR, 2006)

Though the presence of advanced Medicare treatment and better curative means are present to extend the survival rate and enhance the quality of life to the cancer patient by 83%. (Shanta, 1996) This large number of cancer cases has increased the number of new treatment centres two-fold in the last decade. Based on the current trends of cancer occurrence in India, the ICMR (2006) report projects that cancer occurrence would be on the rise from 3.78 million in 1990 (40.4% of all deaths) to 7.63 million in 2020 (66.7% of all deaths) and the NCRP (1999) reports the increase in cancer case in the present decade is high with more cases within the urban population. In this scenario, it becomes critical to design such spaces in a way that it employs best possible results in terms of treatment and healing.

The goal of a healing environment is to provide non-institutional surroundings and a sense of calmness for patients, staff and visitors. Architecture and design can promote the healing process by giving the patients a psychological and physical lift. Healthcare buildings need to actively contribute to the healing process by relaxing and reinvigorating patients.

CONTENTS

- Project Brief
- The Research Process
 - Size and type of the hospital
 - Space and standards
 - Site Entrance
 - Parking
 - Entrance Foyer
 - Reception
 - Corridors
 - Administration
 - Accident and Emergency (A&E)
 - Out-Patient Department (OPD)
 - Diagnosis Department
 - Diagnostic X-Ray Suite
 - Nuclear medicine and Gamma Camera
 - Laboratory
 - Radiotherapy Unit
 - Surgical Suite
 - In-Patient Bed (Wards)
 - Space Design
 - Daylight
 - Furniture Standards
 - Toilets
 - Utilities
 - Portable Bedside Panels
 - Forms and Ventilation
 - High or Low
 - Horizontal
 - Central Large Courtyard
 - Integrated Small Courtyard
 - External Small Courtyard
 - External Small Courtyard
 - Central Large Courtyard
 - Solid Form
 - External Large Courtyard
 - Inference
 - Study of Facade and Shading Devices
 - Ribbon Window
 - Glass Façade

- Skin
 - Louvers
 - Inference
- Growth and Change
- Hospital Street
- Proximity Chart
- Flow Chart
- Standards
- Design Development
 - Site
 - Regulatory Factors
- Site Analysis
 - Size
 - Macro Level Analysis
 - Site Surrounding
 - Vehicular and Pedestrian
 - Climatic Analysis
 - Sun Path and Wind Direction
 - Vegetation
 - Site Slope
 - Drainage Points
 - Impact Generators
 - Access Points
 - Micro Level Analysis
 - Land Use Map
 - Site Access Map
 - Mobility Map
 - Building Height and Infrastructure map
- Room Program
- Case Study
 - Madras medical College Hospital, Chennai
 - Adayar Cancer Institute, Chennai
 - Vedanta Cancer Centre, Raipur
 - Winder Regional Cancer Centre, Ontario
- The Design Process
- Bibliography

PROJECT BRIEF

The forces behind the cancer health care centre are the Indian Institute of Head and Neck Oncology. The centre works in the field of cancer, reaching out to people in re-mote, rural areas of the region, creating awareness on the importance of early detection and the need to get the right treatment and in all this maintain a special focus on oral cancer, given their high incidence in India. The Indore Cancer Foundation works in a small hospital building with only treatment department which provides therapy for curing cancer. They plan to expand it into a specialist cancer care treatment hospital which will serve cancer patients. All successful hospitals, without exception, are based upon a triad good planning, good design and construction and good administration. It is the architect's responsibility to translate clinical administrative needs into architectural and engineering, to integrate different departments of the hospital in such a way that different types of traffic flow in the building are as much segregated as possible, traffic routes are kept short and important functions are protected from prying eyes or intrusion. The skill with which the circulation is handled determines the efficiency of the hospital during its life.

THE IMPACT OF CANCER

Cancer is a widespread and often feared disease because it can affect all people, old as young, healthy as unhealthy and because of the often harsh treatment course and the sometimes deadly outcome. Consequently, being diagnosed and living with cancer bring many unpleasant psychological emotions and reactions alongside physical side effects from the treatment and illness.

It is a dreadful disease that affects millions of people all over the world. Cancer patients spend a lot of time at hospitals depending on the severity of their disease.

In India it is estimated that there are 2 to 2.5 million cancer patients at any given point of time with about 0.7 million new cases coming every year and nearly half die every year. Two-third of the new cancers is presented in advance and incurable stage at the time of diagnosis.

The World Health Organization has said that there could be 50 million new cancer patients by 2020. In India there are 30 lakh cancer patients (till year 2013), the Government hospitals that have cancer treatment facilities tend to pay no attention to the quality of spaces and the impact it makes to the human psychology.

THE RESEARCH PROCESS

SIZE AND TYPE OF THE HOSPITAL

Health services require a wide range of buildings of very different sizes and types. Hospitals may be sub divided into the following categories: smallest (up to 50 beds), small (up to 150 beds), standard (up to 600 beds) and large hospitals (more than 600 beds). Very few of the smallest and the large category hospitals have been built in recent times, the trend now seeming to be to create an even coverage of small and standard hospitals. In fact, modern health reforms have produced a noticeable reduction in the numbers of the smallest hospital. Hospitals are divided by function into general, specialist and university hospitals.

General hospitals: A general hospital provides a range of medical services including: consultations, treatment, diagnosis, minor surgery and health education. Sometimes it may also include day care for physiotherapy and occupational therapy and out-patients emergency treatment. In some cases there may be in-patient short stay beds. These centres can offer great flexibility and tend to serve an average population of 10,000 to 30,000 people.

Specialist hospitals: The number of specialist hospital is growing fast because of the increasing focus on individual types of treatment or medical fields: causality, rehabilitation, allergies, orthopaedics, gynaecology etc. also included in this category are special clinics dealing with, for example, cancers, skin problems, lung conditions, psychiatric disorders, and the like.

University hospitals: University hospitals with maximum provision are to be considered equal to the medical academies and some large general hospitals. They have at their disposals particularly extensive diagnostic and therapeutic facilities and systematically carry out research and teaching. Lecture theatres and demonstration should be included in such a way that operations are not interrupted by the observers.

SPACE AND STANDARDS

SITE ENTRANCES

There should be three prominent entrances to the site:

Main entrance: This is the main entrance for visitors, doctors and hospital staff. It should be the widest entrance as it will be used by maximum people approaching the hospital. The pathway should clearly lead to entrance foyer and parking. **Emergency entrance:** This entrance will be exclusively reserved for the ambulance and emergency cases coming to the hospital. **Service entrance:** There should be a separate entrance for services required in the hospital. This will not disturb the main traffic flow in the hospital; moreover some services should be kept hidden from the visitors.

The pathways from these entrances should not intersect each other as it will disturb the function of each entrance. These pathways should have multiple lanes to maintain continuous flow of traffic.

PARKING

A reserved parking should be provided for doctors and other hospital staff. This reserved parking can be separated from the main visitor's parking or it can also be placed away at a different location on the site. Parking space for public transport such as taxis can also be provided. This will be convenient for the visitors using public transport.

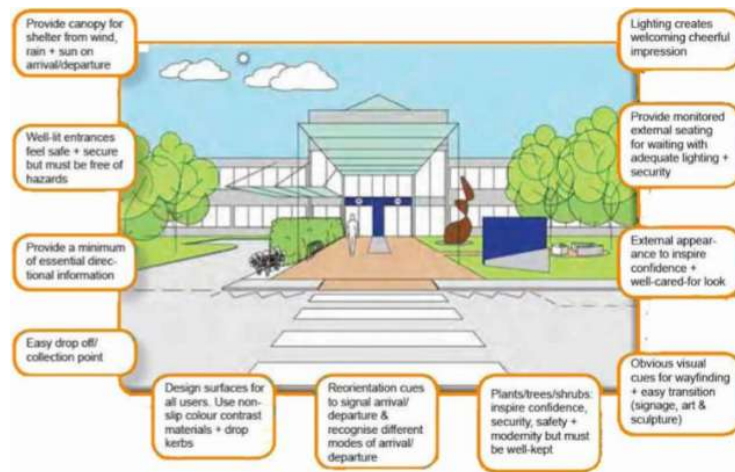
ENTRANCES FOYER

Introduction of character to the main entrance helps people identify it more readily and can create a sense of uniqueness, friendliness and individuality for buildings that often seem institutional or faceless. High and generous entrance ways feel welcoming and uplifting. Low or narrow entrances are claustrophobic and oppressive, not easily identified and can cause uncertainty. There should be uninterrupted and clear views of the entrance from the approach to the site.

RECEPTION

The reception, information or help desk should be immediately apparent but not prevent people seeing the rest of the space or become an obstacle in itself. Being able to see shops, cafés, toilets, cash machines and other facilities from the entrance makes people aware of them and reduces the need for signs. Being able to see staircases

and lifts from the entrance helps people find their way more quickly.



CORRIDORS

Recessed seating areas along corridors provide places for patients and staff to meet and rest. Seating that coincides with views from corridors to the outside offers beneficial respite for patients, staff and visitors. Typical routes along corridors can be suggested by materials and colours on walls and floors. Projecting signs are easier to see when walking down a corridor than signs fixed flat against the wall. Creating landmarks with art, sculptures and design features helps navigation. Views from circulating spaces into bed and patient spaces should be limited.

ADMINISTRATION

The extent of offices provided in the hospital depends on whether they are also the headquarters of the Trust. Main functions include Trust Board and Secretariat, finance, personnel, supplies, and senior nursing and other professional staff. Ancillary spaces include computer facilities, stationery and other stores. The reception desk and enquiry desk should be associated with the main entrance. Other administrative functions do not need priority locations, and may be on an upper floor.

ACCIDENT AND EMERGENCY (A&E)

The accident and emergency department should be near the hospital entrance. It should be in a good proximity with the main entrance, the pharmacy and reception and enquiry desk. Its access should be through a ramp and there should be enough space outside the department where the ambulance can be parked. This department should have a lift specifically for critical cases. It will also have a minor OT,

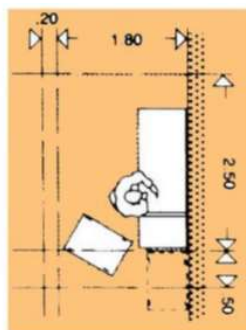
causality, and a doctor's cabin, who will supervise the department.

OUT-PATIENT DEPARTMENT (OPD)

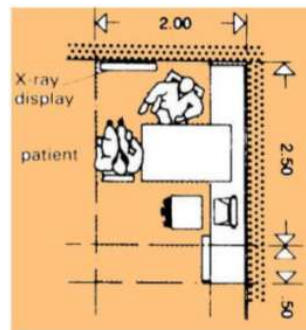
The contracting out of services following health reforms has freed space in many hospitals. Much of these has been converted into day clinics for patients who are only cared for during the day and do not require hospital beds, or who have undergone outpatient surgery. As these patients are divorced from the rest of the hospital activities it is



necessary to provide a separate entrance to them. Out-patient surgeries for minor operations are becoming increasingly common. In a hospital, the outpatient surgery should be close to emergency room and surgical department. The sequence for the patient is reception, registration, waiting and consultation. A general waiting area should be provided for family with a refreshment counter and a children's play room. It should have a close proximity with radio diagnostic facilities, pharmacy, physiotherapy (if possible).



Minimum area: examination of reclining patient



Minimum area: doctor's consultation

STANDARDS FOR CONSULTATION ROOM IN OPD

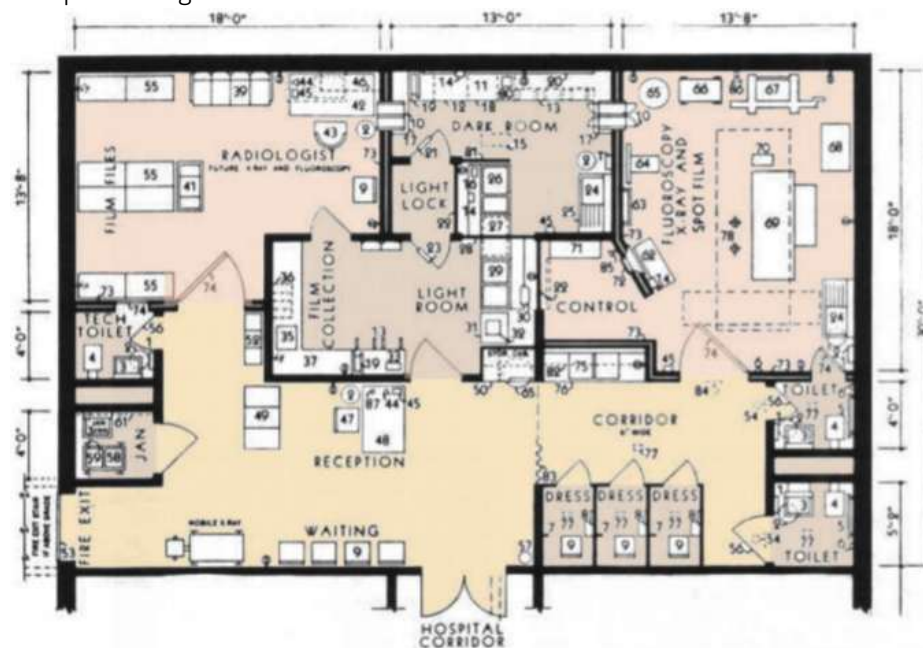
DIAGNOSYS DEPARTMENT

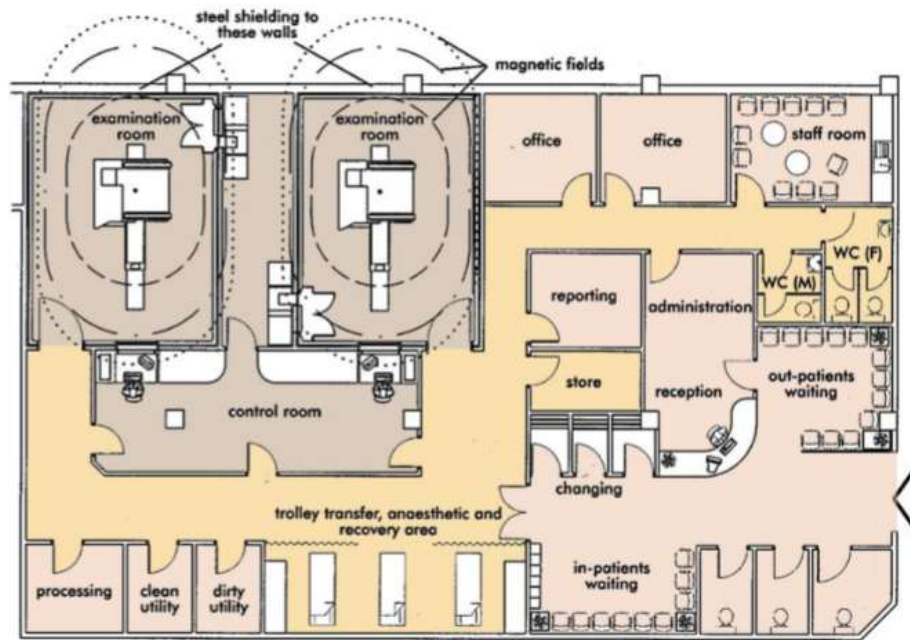
All the diagnosis units like X-ray, CT scan, and MRI etc. falls under this head. There are standards which have to follow while designing these units as some of them may radiate harmful radiations.

The UNITS's are-

DIAGNOSTIC X-RAY SUITE

The diagnostic x-ray department should be located at the first floor, conveniently accessible to both in-patients and out-patients. It is desirable to locate the departments close to the elevator and adjoining the outpatient department and near other diagnostic and treatment facilities. The staff needed for this department includes: 1 radiologist, 2 or 3 technicians, 1 secretary receptionist 1 clerk. Darkroom is located between the two x-ray rooms to facilitate handling of films. The films are loaded and unloaded on the counter. Space is provided for loading and stacking films at both ends of the counter. A light lock between the darkroom and light room equipped with interlocking door is necessary to allow entrance into the darkroom of other personnel during film processing.

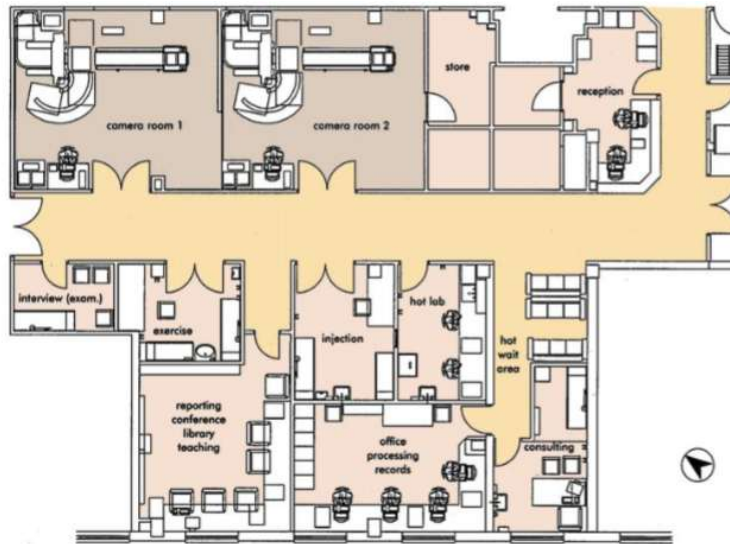




NUCLEAR MEDICINE AND GAMMA CAMERA

The treatment of patients with radiotherapy mostly, but not exclusively, cancer patients - requires the specific provision of protected facilities. The general departmental requirements are not dissimilar to those of diagnostic imaging departments, with the need for reassuring reception and waiting for spaces and, as all the treatment rooms are internal, for as much of the general departmental spaces to have daylight and outlook. With the more complex modalities, such as linear accelerators, the physics requirements are highly specific and complex, such that these very high powered machines usually are given their protection by housing them in a massive concrete bunker with maze entries to prevent irradiation leakage. Nuclear medicine involves the use of radioisotopes in the diagnostic or therapeutic mode.

The process embraces:



The preparation of the isotopes their injection into the patient tracking them in the body Measuring the rate and degree of their uptake using gamma cameras.

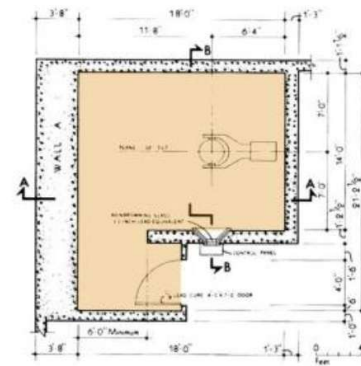
Therefore a nuclear medicine department must contain a radiation protected isotope store, a laboratory for manipulation and preparation of the isotopes, and a gamma camera room or rooms where the procedures will take place.

LABORATORY

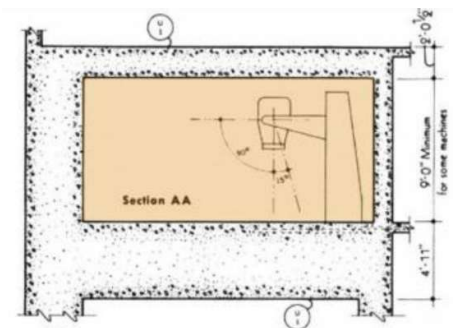
Locate the department as favourably as possible for the laboratory staff and the ambulant inpatients and outpatients. A space on the first floor near an elevator is preferable. In determining the overall size of the laboratory, the first concern is the individual technical units. It is only after the size of these units has been established and an architectural layout has been developed to fit the program that the sum of the areas can accurately reflect the size of the laboratory department. The laboratory services of a general hospital would require work areas for six main technical units: Haematology, blood bank, urinalysis, biochemistry, histology serology and bacteriology.

RADIOTHERAPY UNIT

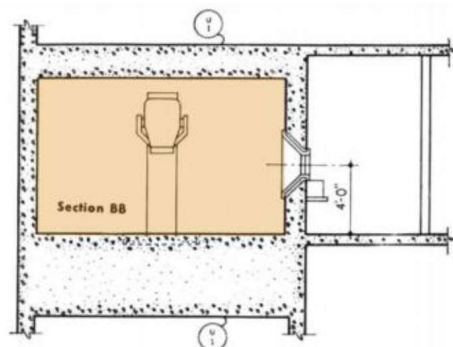
In radiotherapy, conditions diagnosed in the diagnosis department (e.g. tumours) are treated. The different rooms which support the therapy units are reception area, console room, the machine room, UPS room, TPS room (Treatment Planning System) and store. The room size may vary to suit different manufactures equipment. A room approximately 4500mm to 5500mm plus necessary entrance maze will accommodate most of the machines commercially available. Shielding is very important with this unit as the rays are very harmful for humans. Therefore thick concrete walls are made and these units are planned in the basement of the building so that the radiations gets absorbed by the earth and architects should try not to place any other function beside this. The walls, floor and the ceiling are all made up of concrete. The shielding necessary should also be considered for doors and other openings. In designing the door of such room, a good practice is to have a door of wood with a layer of lead on it.



Cobalt-60, treatment room: Plan



Section A-A



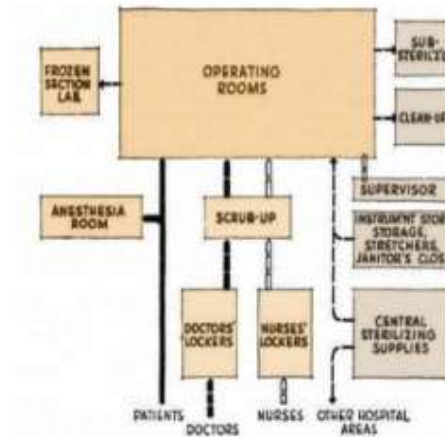
Section B-B

SURGICAL SUITE

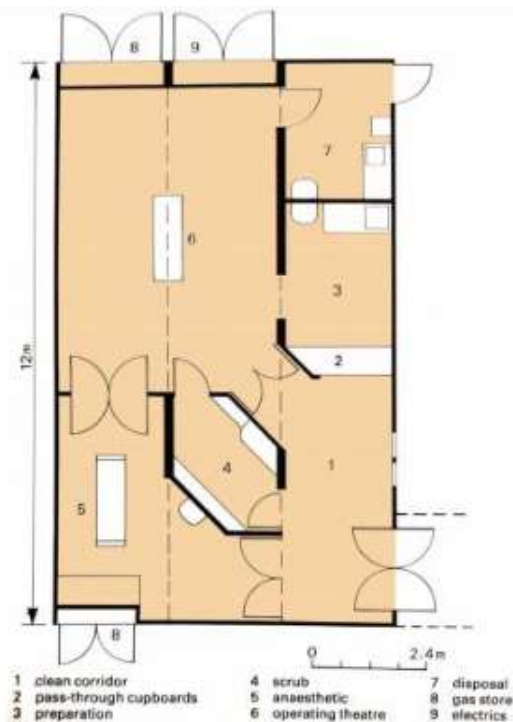
The surgical suite of the hospital is a very complex workshop. There may be different number and different type of operating rooms. In general hospital all major operating rooms are identical as to facilitate scheduling of various surgical procedures. The planning and equipping of each operating room is based on a series of questions, such as: size, usage, lighting, intercommunication, electric equipment and monitoring system, service line such as suction, oxygen, nitrous oxide and air, provision for TV cameras, safely precautions in hazardous areas and storage cabinets. The average size of the operation theatre is around 40 sq. m.

Administration area, scrub area, work and supply room, laboratory, post-aesthetic recovery, lounge, locker and toilets for various personal groups and circulation within the department. Within the surgical units we have three basic zone predicted on three types of activity and circulation involved and degree of sterility to be maintained. Outer zone: Administrative elements and basic control where personnel enters the department, patients are received and held or send to proper holding areas of inner zones; conference room, locker spaces, any outpatient reception etc. Intermediate zone: Predominantly work and storage areas; outside personnel will deliver to this area but should not penetrate the inner zone. The recovery unit if integrated with surgical suite, is an intermediate or outer zone activity. Inner zone: The actual operating room the scrub area, the patient holding or induction areas. All alien traffic should be eliminated. Here we want to maintain the highest level of cleanliness and septic conditions.

The surgical suite location must mesh with the total circulation pattern so that the patients can be moved to and from surgery within a minimum of travel through other hospital services. The Intensive Care Unit (ICU) is generally placed close to the operation theatre as the patient who has just gone through a surgery is under direct supervision of the doctors and nurses. After the doctors feel that the condition of the patient is

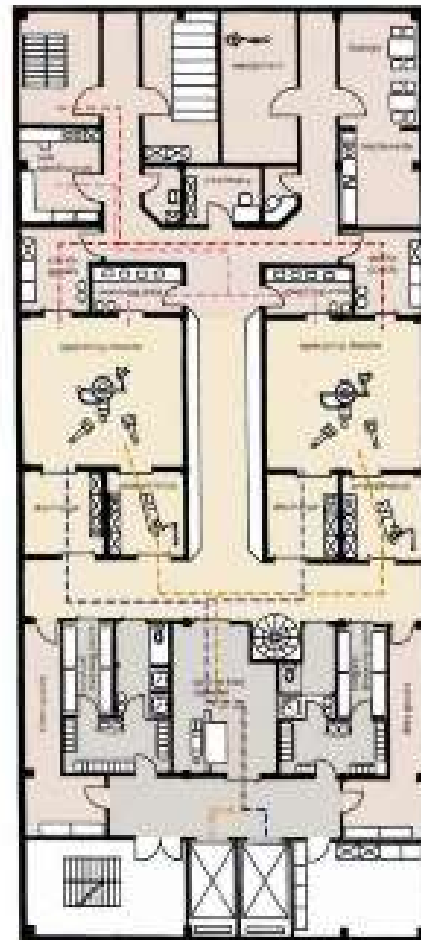


Circulation in OT Suite

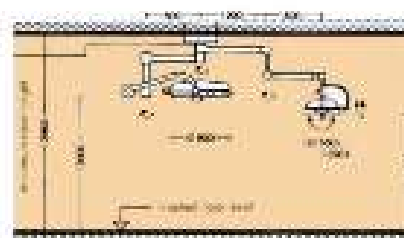


OT suite by Trojan structures limited. Three standard 12X2.4m ISO containers arranged side by side. The unit includes a flexible corridor link for connection to existing buildings and a roof mounted air conditioning package. Electrical switchgear and medical gases are accessible from outside the unit.

stable they transfer them to their wards or rooms. The air conditioning in the OT is different as in other areas. It is centrally air conditioned in which purified air enters through ceiling and after some time it gets sucked by the vents present at the bottom of the walls. 15-20 air changes per hour are required to ensure adequate decontamination of the air between operations.



- Patient's Circulation
- Doctor's Circulation
- Utilities from Central Sterilization
- Utilities from Sub-Sterilization
- Circulation in OI Suite



Operation theater Section

IN-PATIENT BEDS (WARDS)

The patients who live in hospital while under treatment or the patients who need all-time attention of hospital staff and access to hospital services are admitted in rooms or wards. Broadly they are further classified as Private rooms: These are single-bedded rooms with an attached toilet and another bed for the visitor, other facilities like dining space, lounge etc. increase with the increase in tariff rates. Semi-private rooms: These rooms may be double-bedded or four-bedded with a common toilet and a bed for visitor with each patient bed. General ward: Wards are big and long rooms with a preferred range of 20-30 beds. It may have one or two toilets which will be shared by all. They are categorized as male and female wards. In some places we may also find mixed-sex wards.

SPACE DESIGN

Creating zones for patients, visitors and care takers within the bed place helps each feel a greater sense of ownership and belonging. Providing opportunities for displaying pictures and other personal possessions is important. Visual and audible privacy for patients undergoing treatment are crucial to maintaining patient dignity. Furniture, screens and the positioning of beds can create a more personal space in multi-bed rooms.

DAYLIGHT

All bed places should ideally be exposed to daylight. The beds should be placed parallel to the exterior wall, so that the patients can look out of the window without facing directly into the bright sky. Daylight is important for confined patients to maintain a sense of time and natural body rhythms. A lack of daylight will depress confined patients and could add to despondency.

Direct sunlight should be avoided or shaded as it can be uncomfortable and irritating for patients in bed who cannot avoid it. Bedside controls of blinds and curtains help reduce frustration and restore a sense of independence.

FURNITURE STANDARDS

Care takers must have access to at least one side of the bed. Doorways and circulation space must allow for trolleys and wheelchairs. The size of the adjustable bed is 2200mm by 610 mm with an adjustable height of 550mm to 900mm. The typical bedside table measures 400mm to 500mm. The wardrobe of size 550mm by 2100mm and 550mm deep is sufficient (this size may vary).

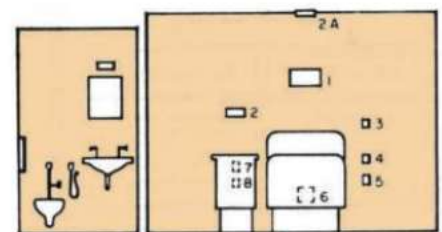
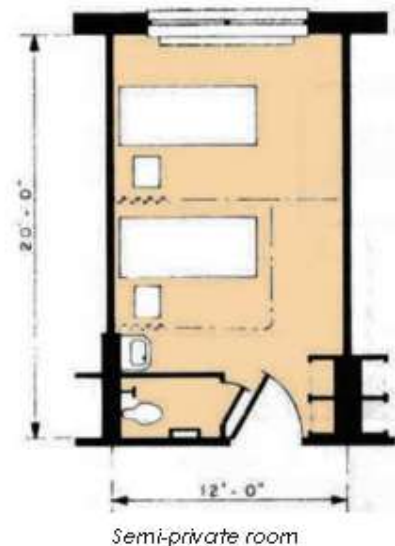
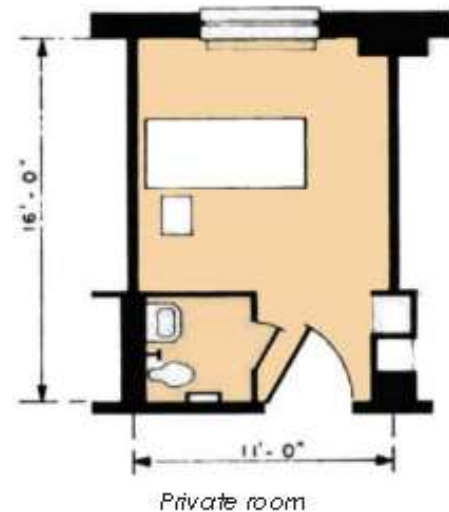
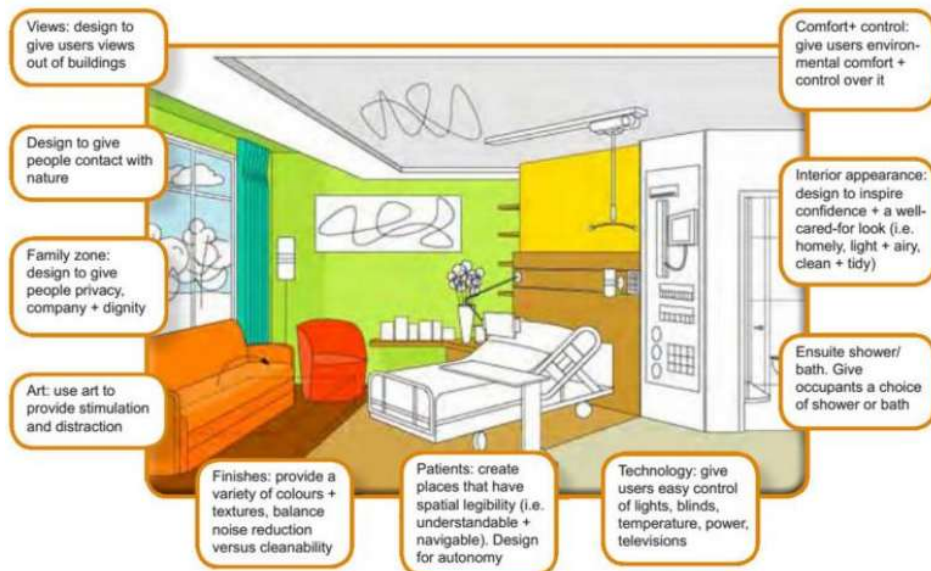


Fig. 7. Wall elevation of single room (left) and double room (right)

Legend

- | | |
|------------------------------|---------------------------|
| 1. Overbed light | 3. Oxygen outlet |
| 2. Nurses' call | 4. Suction outlet |
| 2A. Micro speaker in ceiling | 5. Suction bottle bracket |

Furniture standards in rooms



TOILETS

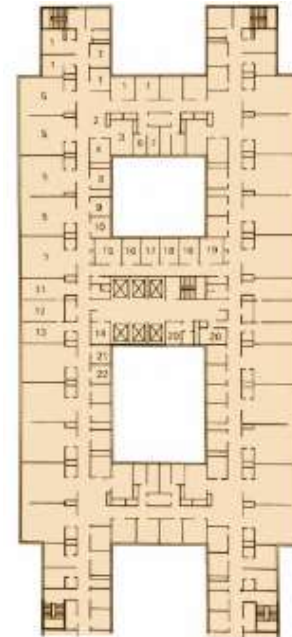
1500mm by 2500mm are the dimensions noted for individual -toilets. The toilet door swing out into the bedroom, except where surface bolted or pivoted hinges are used, so that the door can be removed in the event a patient in toilet faints and fall against door.

UTILITIES

The oxygen is piped in from a central source in most rooms. Outlets are 1200mm to 1600mm above floor level. The suction outlets are either grouped in same plate with oxygen or they are separate, beside or below oxygen.

PORTABLE BEDSIDE PANEL

Nurse call switch, pilot light Monitor light Reading light switch Room thermostat remote control Electric clock Provision for TV remote control Provision for telephone instrument



1 single bed	9 kitchen	17 medical x-ray
2 nurses station	10 toilet	18 nursing supervisor
3 broomroom	11 day room	19 laboratory
4 clean utility	12 dining	20 x-ray overnight
5 free bed ward	13 TV	21 secretary
6 dirty utility	14 reception	22 cleaners
7 bath	15 doctors on-call	
8 storage/ward	16 consultant	

Half a ward floor at Yerk General Hospital, comprising four 30-bed nursing units, each with one nurses' station. Each unit has five single-rooms with five multi-bed bays of five beds; the single rooms in the cross links are 'swing beds' and maybe used by either of the two flanking units as required. Day, dining and TV rooms are shared between adjacent units. The ward building is four stories in height and is naturally ventilated. Architects: Llewellyn-Davies Weeks

FORM AND VENTILATION

Looking at the examples of hospitals, the form and the position of the buildings are an important aspect. The form can influence the comfort of the user. The size and complexity of hospital can be daunting to staff and visitors, and can be overwhelming to ill and anxious patients. Simplicity of access to and between the areas generating most traffic is essential. Although the fire risk in the hospitals are low means of escape criteria and stringent. This is because of the high proportion of people who are unfamiliar with the detailed building layout and include the bed-bound, sick and those with disabilities. It is good advice that "the way-in should also generally be the way-out": in other words, evacuation will be most effective if it is by well-known and well-used routes. There may well be conflict between planning in a form loose enough to provide for future change and expansion and the distance that staff and patients need to travel to get from one place to another.

HIGH OR LOW?

The taller the building, the more inherently expensive, especially as the proportion of floor space occupied by stairs, lift and shafts increases with height, as does the need for total artificial ventilation. Lately, the lifts in some tall hospitals have proved inadequate to cope with the greater volume of vertical travel caused by shorter and more intensive in-patient stays. Horizontal contiguity of functionally related departments has been a guiding principle. That is, functionally related departments should be if possible close to, and on the same level at each other. This avoids time wastage, especially when moving supplies, equipment, and patients on trolleys and in wheelchairs. Also, to be fairly close to the ground, to open a window, to smell the world outside may reassure the patient that he is still connected with the outside world. I established for me a system of broadly two type namely horizontal forms and vertical forms. These are further classified according to different plan forms. This system helped me to analyse the advantages and disadvantages of different forms and the influence of the form on the use of the building. The forms are based mostly on simple shapes or shapes composed from simple shapes. This is influenced mostly by the fact that simpler solutions may also offer a psychological advantage for staff and patients, particularly for patients who suddenly find themselves in an alien and disturbing world of medical procedures and unnatural interventions.

Horizontal Typology



Central Large Courtyard



Integrated Small Courtyards



Internal and External Small Courtyards



External Small Courtyards

Vertical Typology



Solid



Central Large Courtyard



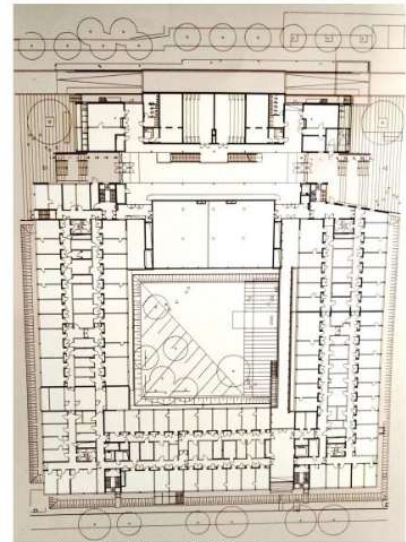
External Large Courtyards

HORIZONTAL

CENTRAL LARGE COURTYARD



A central large courtyard results in a single long main hospital street carrying traffic of all hospital departments. These courtyards are also not used by in-patients due to its public nature which makes it noisy and fails to give privacy to the patients. The form is less flexible for future expansion needs as it is inward looking and have only a single circulation system.

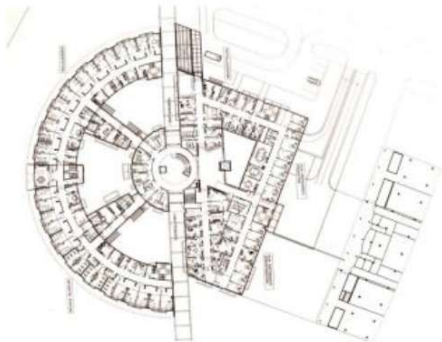


Faculty of Medicine, Carl Gustav
CarusWV Tech. University

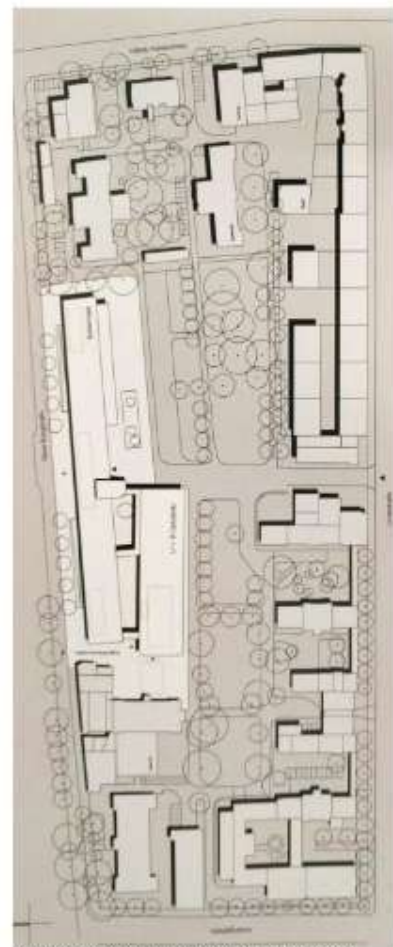
INTEGRATED SMALL COURTYARDS



Small courtyards offer a lot of advantages over a single central courtyard in hospital planning. Small courtyards act as semi-private zone for different wards while a large courtyard is more public and thus less used by patients. This form also separates main hospital street from sub streets of different departments which considerably reduce traffic. Shaded internal courtyards acts as light source for the circulation corridors as well as other hospital areas like wards and waiting areas, where light and plantation in the courtyard pleases the patients and helps in faster recovery.



Vivantes Clinical Center, Neukölln, Berlin.

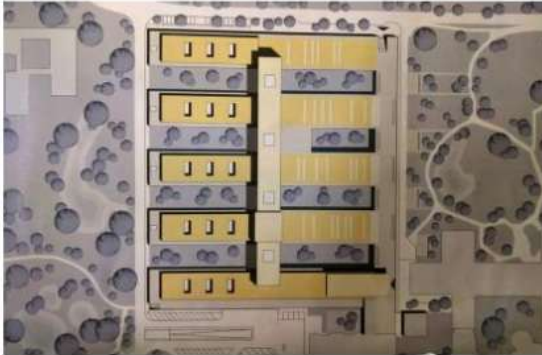


Vivantes Clinical Center, Spandau
Berlin

EXTERNAL SMALL COURTYARDS



This form has a main central spine, from which long fingers enclosing courtyards are developed outwards. The main spine acts as the main hospital street which can be extended farther to provide for future expansion needs, though the presence of single spine elongates the horizontal distance between departments.



Helios Clinical Centre, Berlin-Buch.

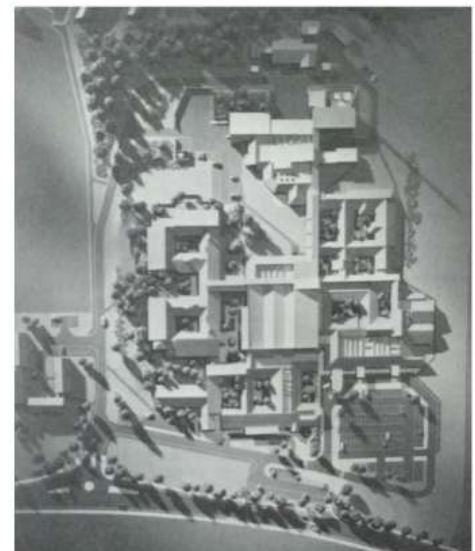


University Centre, Ulm.

EXTERNAL SMALL COURTYARDS

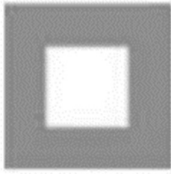


These forms have same features as the integrated small courtyards with some additional features. They further divide sub-street into secondary streets which further distributes traffic flow. These plans also provide courtyard on both the sides which allows for greater depth of rooms and also optimizes cross ventilation.

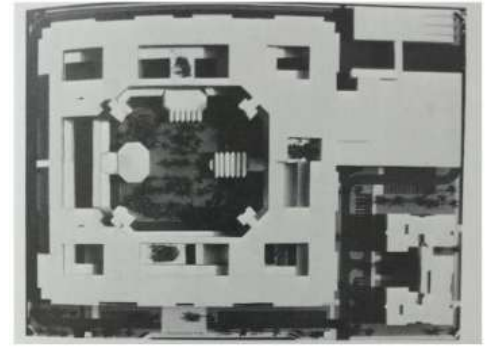


North Tyneside District General Hospital

CENTRAL LARGE COURTYARD

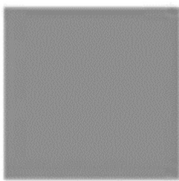


In case of tall hospitals a central large courtyard acts as an advantageous built form. The courtyard helps in optimum ventilation of all departments and acts as a light well for all floors. This type of courtyard planning is well suited for tropical climates.



Al-Karma Hospital, Baghdad

SOLID FORM



Solid tall hospitals have more disadvantages over other forms. Absence of courtyard results in use of more artificial light which consumes more power and it do not give comfortable and pleasing environment for the ill. These types of buildings do not have optimum ventilation and thus rely on artificial ventilation systems which are more expensive.



Apollo Cancer Hospital, Chennai, India.

EXTERNAL LARGE COURTYARD



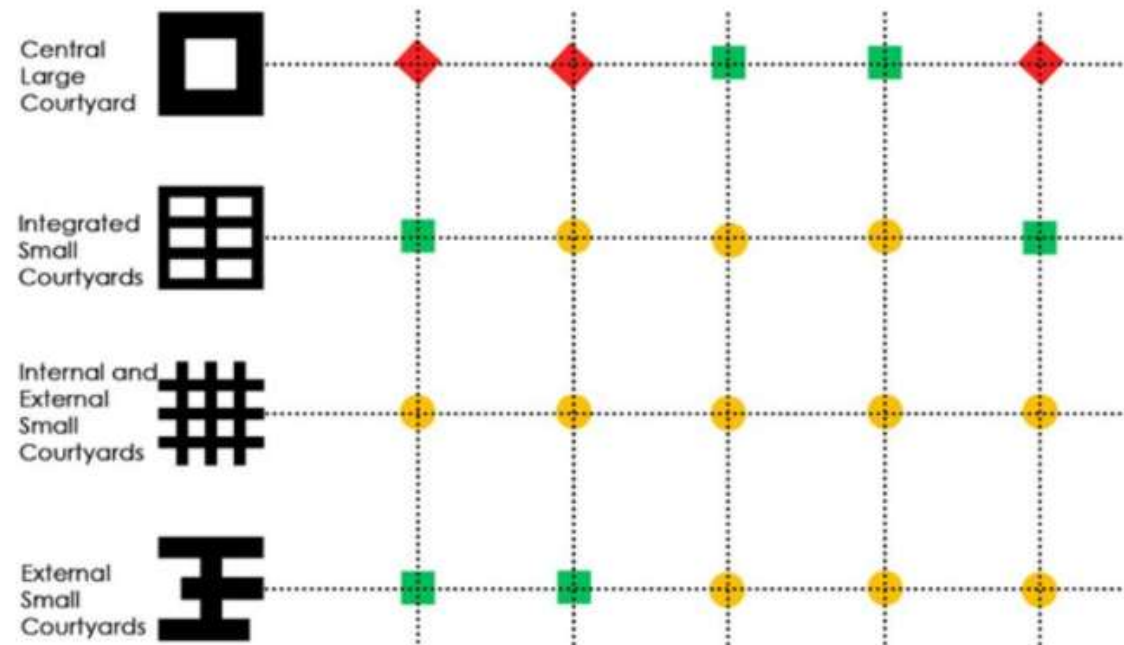
In tall hospitals External large courtyards have same advantages as internal large courtyards. In addition they provide better circulation and separation of departments and a better penetration of natural light into the building if there are multiple external courtyards.



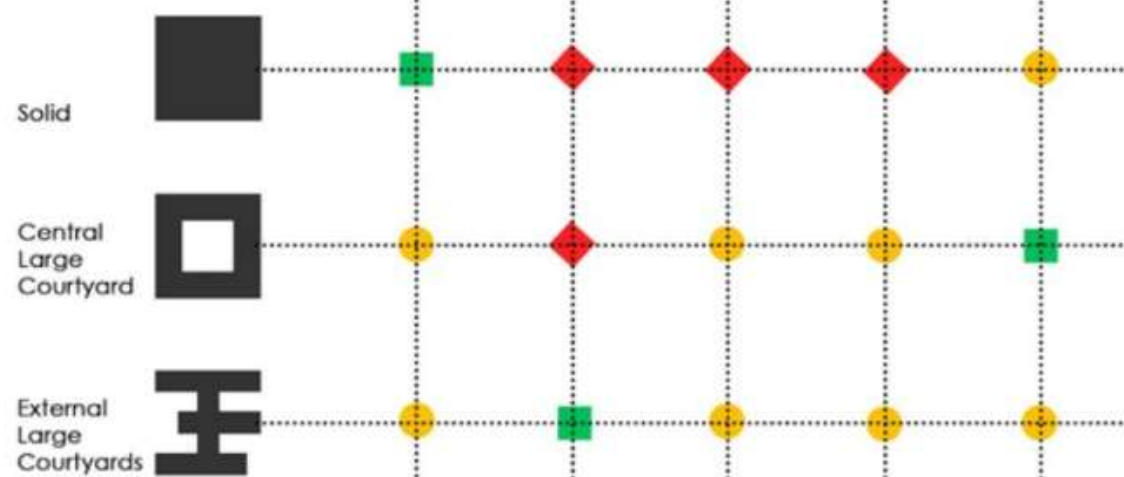
Bombay Hospital, Indore India.

INFERENCE

Horizontal Typology

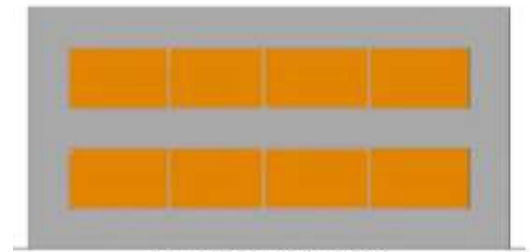


Vertical Typology

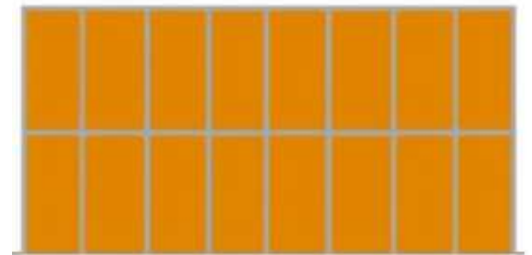


STUDY OF FAÇADE AND SHADING DEVICES

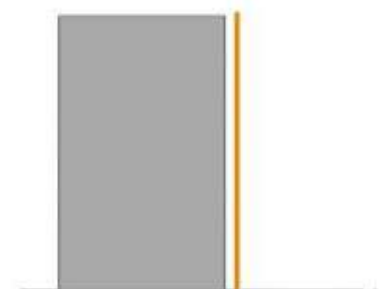
According to my study there are four major types of facade used in hospital buildings. According to my site, i.e. Indore, India which is a tropical zone sun is the major issue. Thus fully glazed facade is not preferred in this region. Ribbon windows is a good facade option as it provides a clear view and it allows optimum light and wind inside the building. The heat gain can be controlled by providing structural shading divides such as push in or pull out systems or external shading devices such as louvers. The advantage of using louvers is that they can be designed according to the need of different building facades which can efficiently manage light and heat gain. It is possible to change the strategy of the shading on different facade fronts as each facade receives sun at different angle with different intensity.



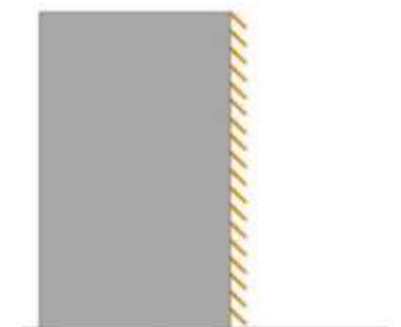
RIBBON WINDOWS



GLASS FACADE

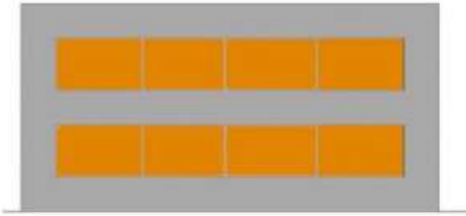


SKIN



LOUVERS

RIBBON WINDOWS



Alb Hospital, Munsing



Krefeld Clinical Center
(Ludes Architects)

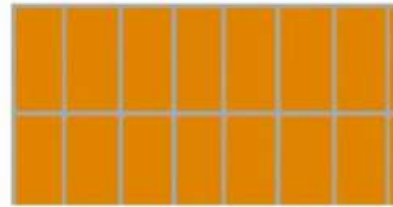


Bruck/Mur State Hospital



Hospital of Holy Spirit, Bingen

GLASS FACADE



Aschau Orthopaedic
Paediatric Clinical



Rosenheim Clinical Center



St. Elizabeth and St. Barbara
Hospital, Halle/Saale



Basel University Hospital

SKIN



Robert Koch Clinical Center,
Leipzig



Clinical Center of Friedrich
Schiller university of Jena



Boblingen District Hospital

LOUVERS



Sophien und Hufeland Clinical
Center



Graz-West State Hospital

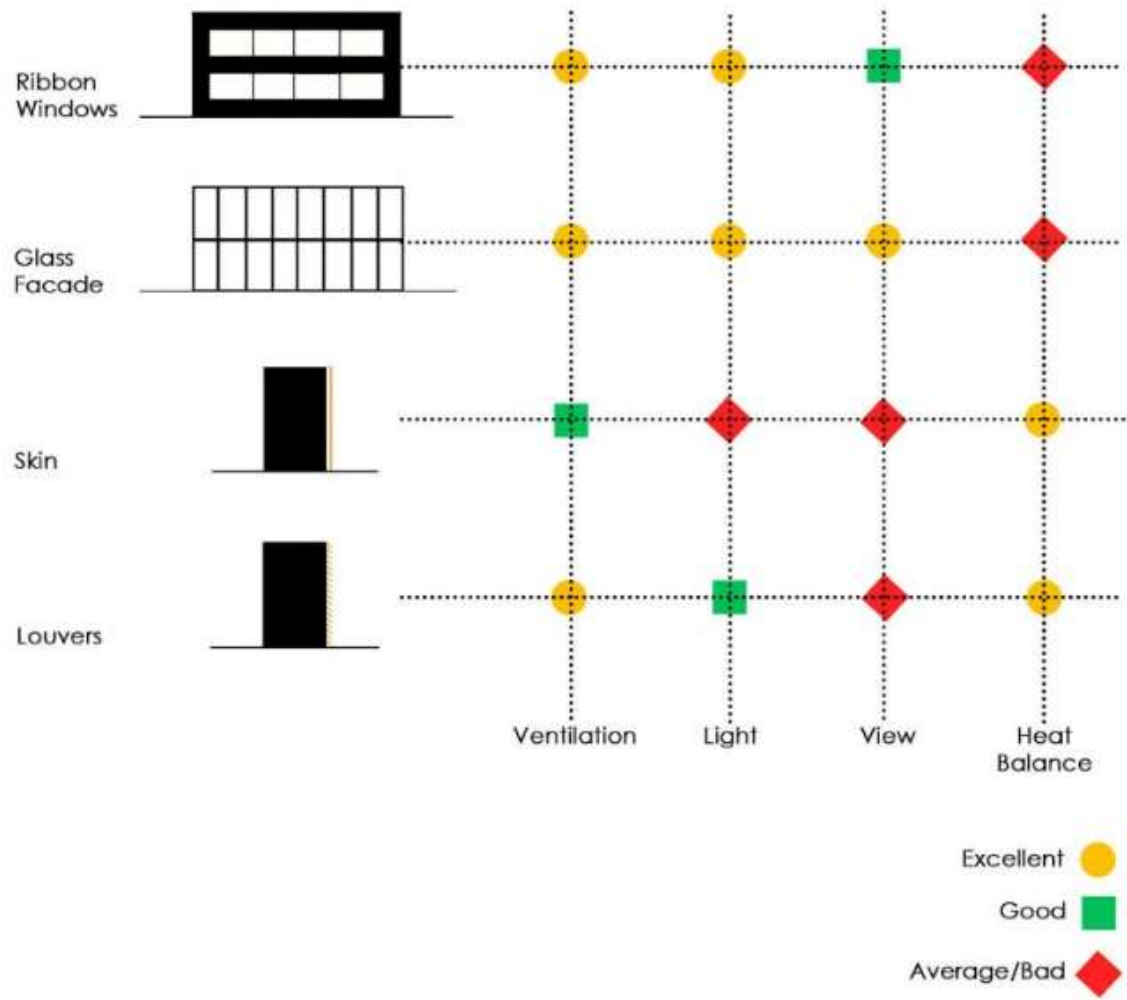


Erding District Hospital



Rehab Basel

INFERENCE



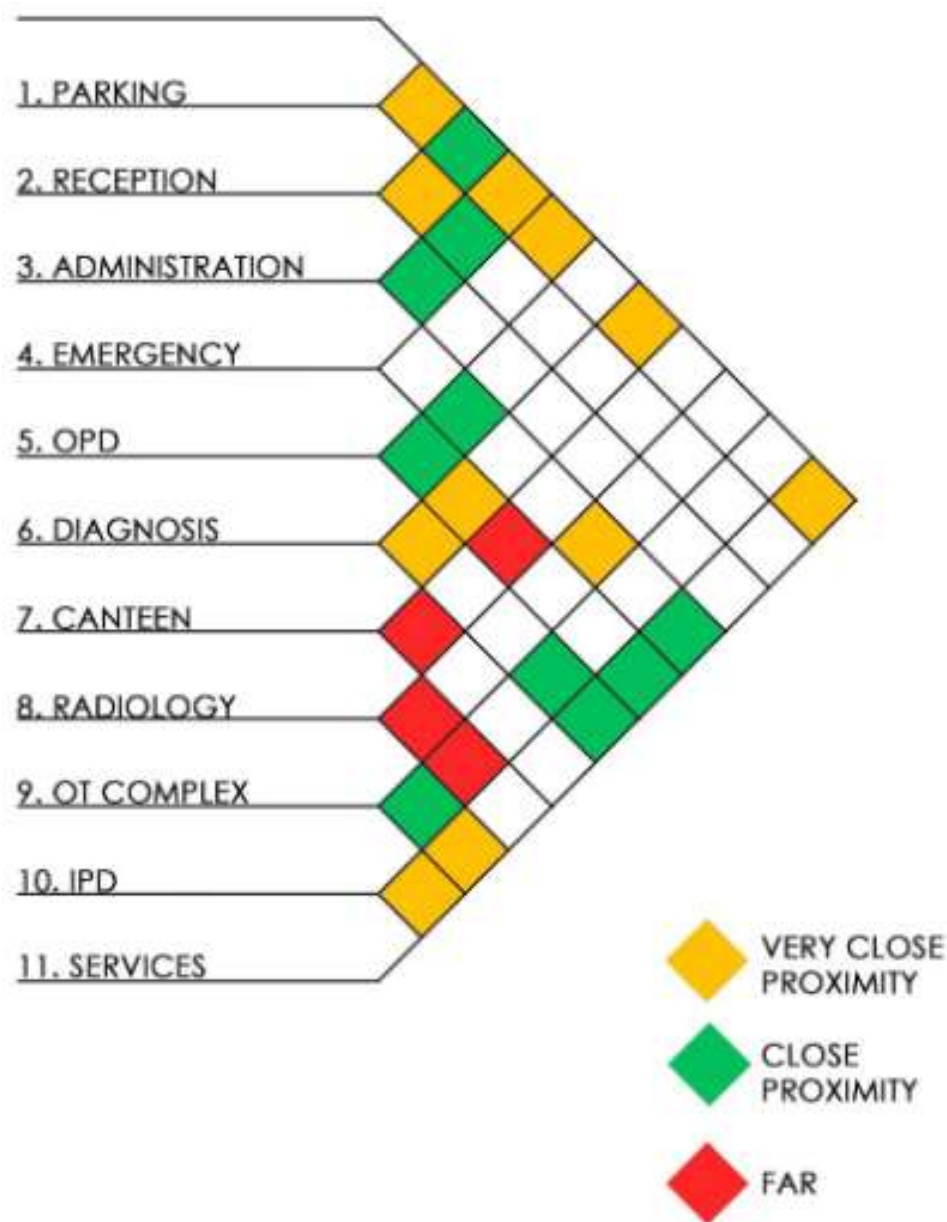
GROWTH AND CHANGE

As the inevitability of growth and change has come to be realized views have been expressed during the last few decades which maintain that, wherever it is situated, a hospital is never likely to be complete. Hospitals are built in several phases or are added in stages to existing hospitals. As the building will inevitably require alterations and additions during its working life, from the outset its physical organization should be conceived with this in mind such as to allow a variety of extension possibilities.

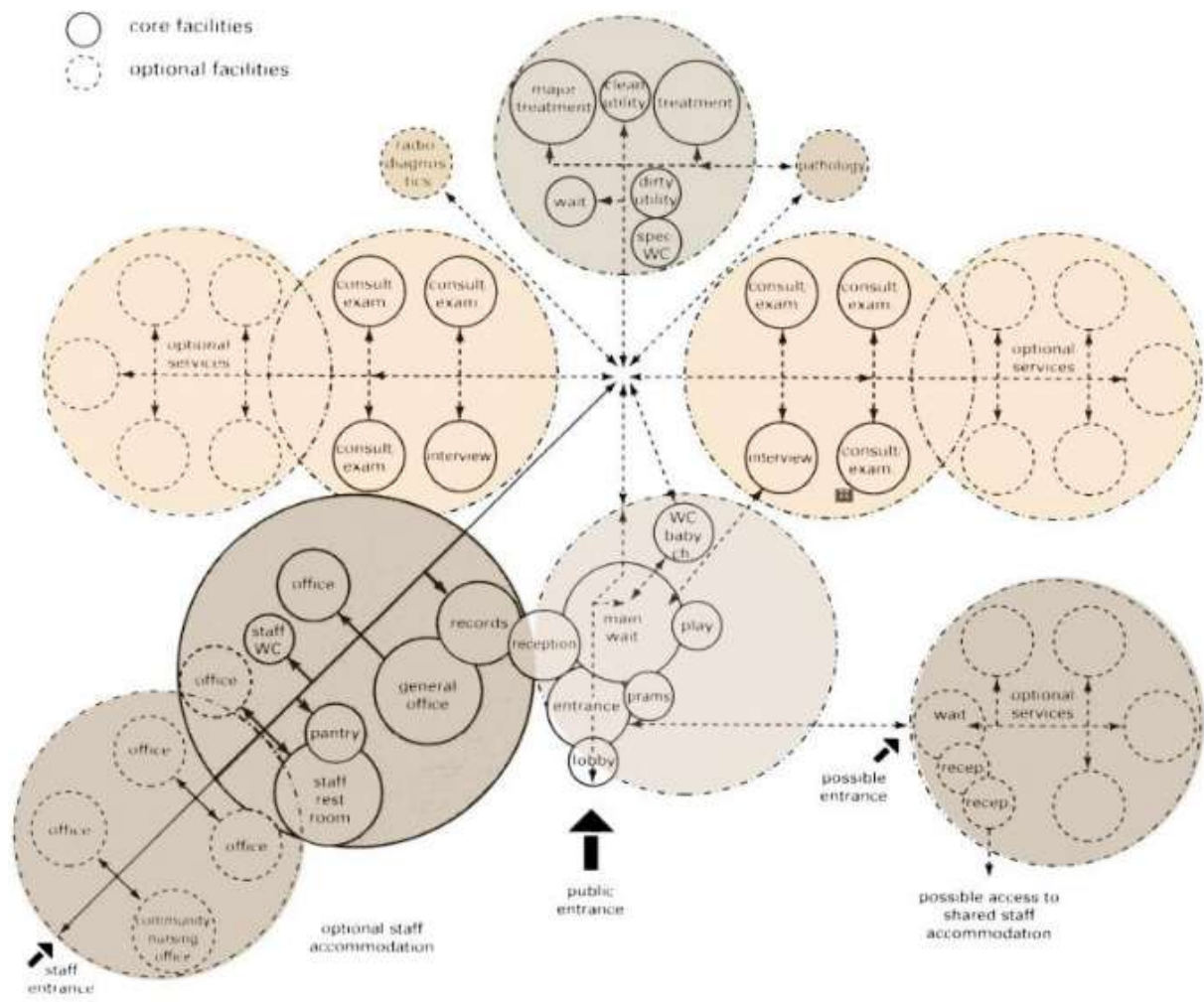
THE HOSPITAL STREET

It is desirable to provide strict segregation between the streets for supplies or other goods and those for people, although by the disposition of departments it may be possible to avoid serious conflict between the major categories of traffic, both from point of view of its density and of potential risk of cross infection. It is advisable to separate the main circulation spine from secondary circulation of smaller departments; this distributes the traffic flow from the main street which reduces chaos in corridor areas. The hospital street can have impact on future growth and expansion. A single continuous spine allows expansion by expansion of the street while a closed loop type street is difficult for future expansions.

PROXIMITY CHART

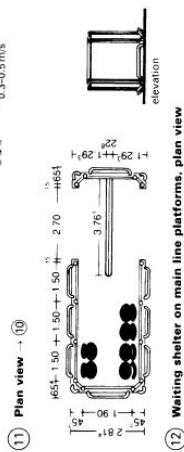
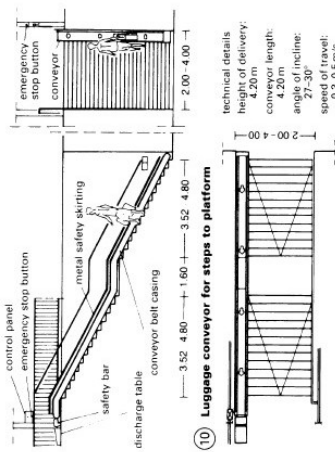
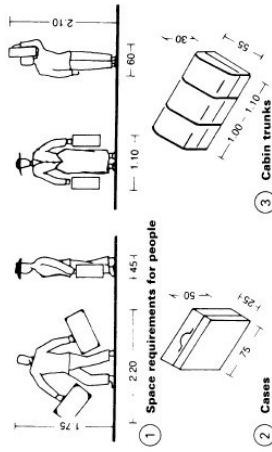


FLOW CHART

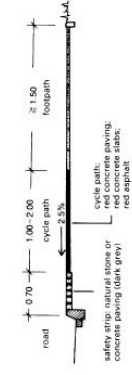
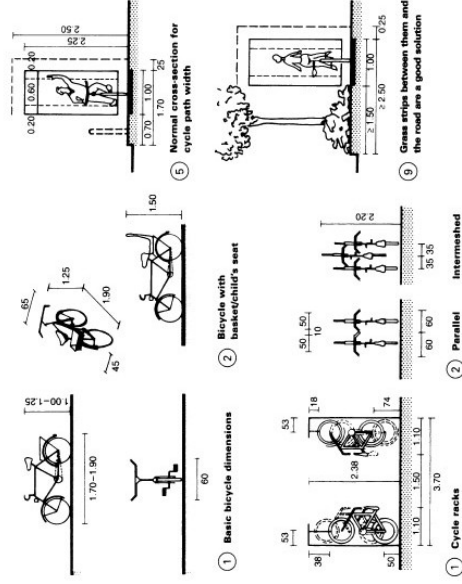


STANDARD'S STUDY:

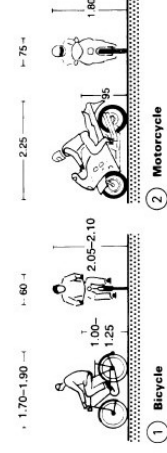
PEDESTRIAN



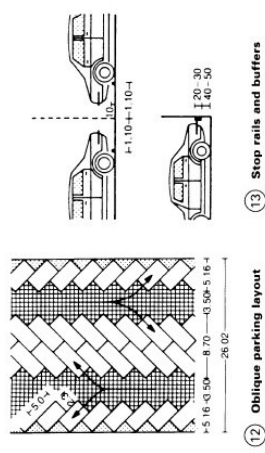
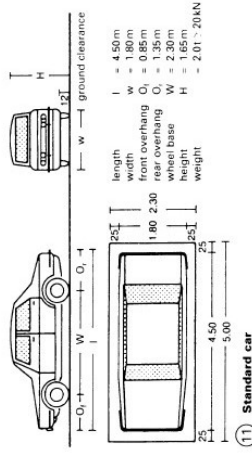
BICYCLE DIMENSIONS



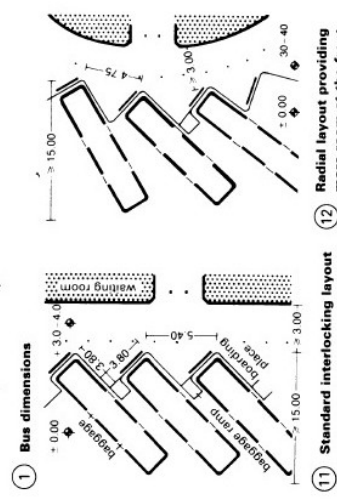
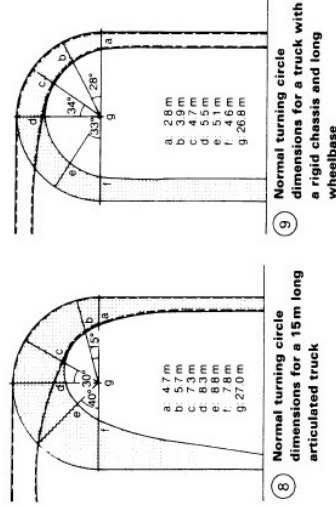
BIKE DIMENSIONS



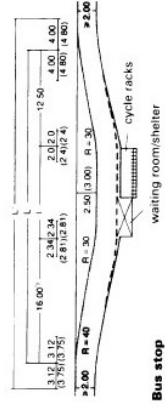
CAR DIMENSIONS



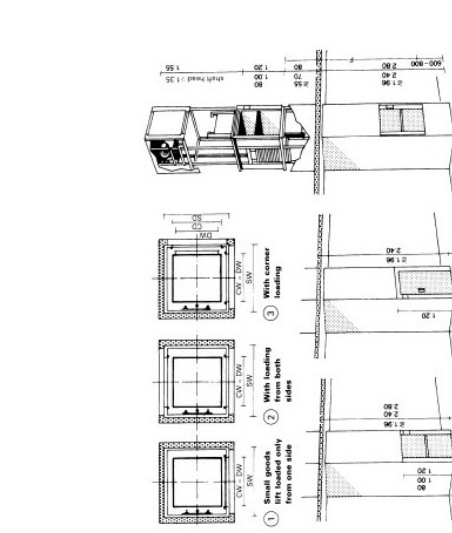
BUS PARKING BAYS



	I	L	L'
bus	12.00	40.50	47.62 (49.05)
two buses	25.00	53.50	60.62 (62.05)
articulated bus	18.00	46.50	53.62 (55.05)



ELEVATOR STANDARDS AND SMALL GOODS ELEVATOR

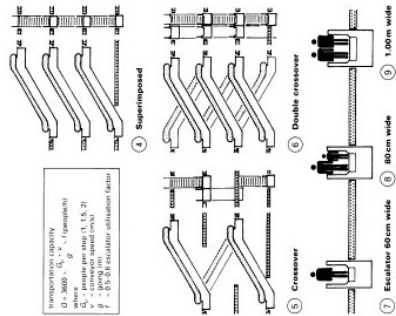


② Escalator width

115

	step width	600	800	1000
A		605-620	805-820	1005-1020
B		1170-1220	1320-1420	1570-1620
C		1280	1480	1680
	transportation capacity	5000-6000 persons	7000-8000 persons	9000-10,000 persons

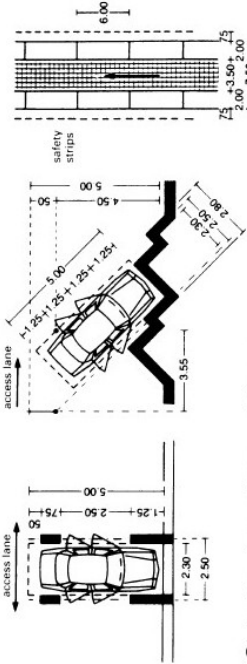
③ Dimensions and performance for escalators with either 30° or 35° angle of ascent



Length in plan (1)

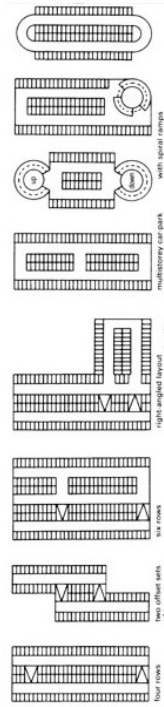
with 30° escalator = $1.732 \times \text{storey height}$
 with 35° escalator = $1.428 \times \text{storey height}$
 Example: storey height 4.50 m and angle 30° (note that 35° angle is not allowed in most countries)
 length in plan = 7.794
 Including landings top and bottom, total length is approximately 9 m, allowing for about 20 people to stand in a row on the escalator.

speed	time per person	width sufficient for:
0.6 m/s	~ 18 s	1 person
0.65 m/s	~ 14 s	2 persons



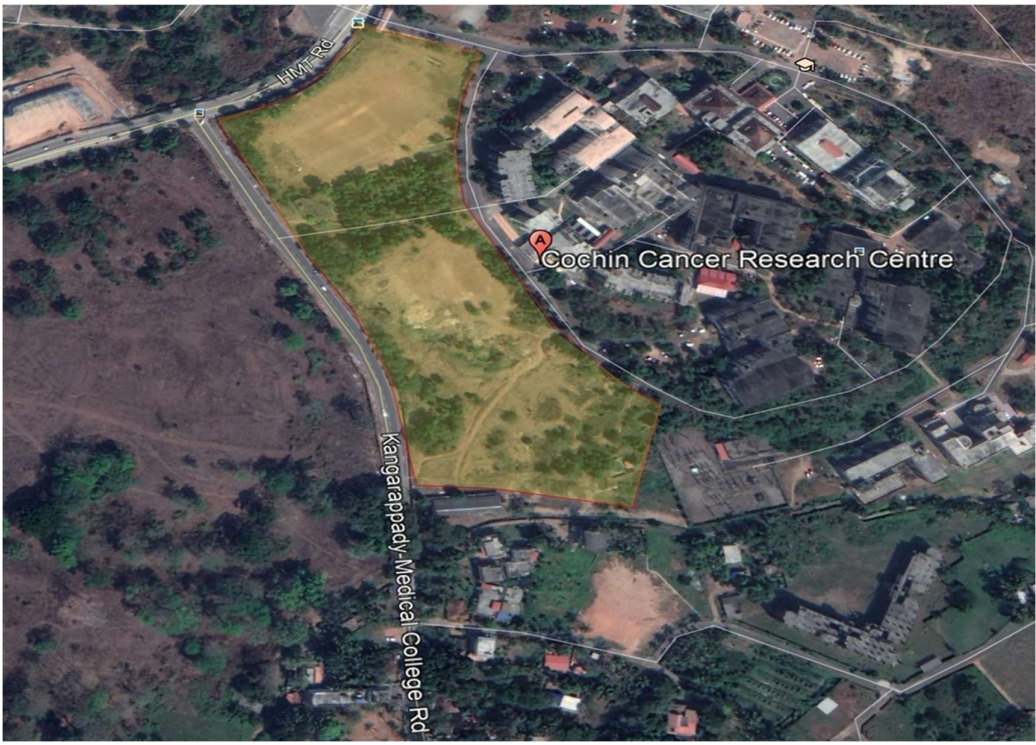
① Possible column positions, parking at 90° to access lane

② Parking at 45° to access lane)



⑦ **Plan views and arrangement of ramps**
of two rows

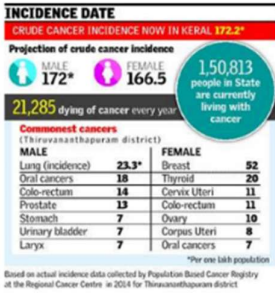
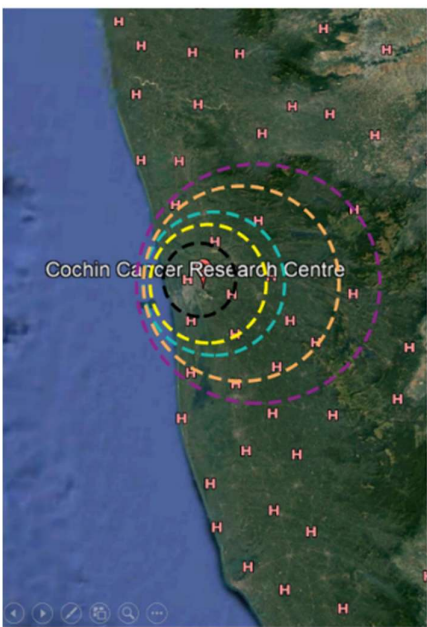
DESIGN DEVELOPMENT



Site Location: Kalamassery, Cochin
Site Area : 14Acres

The site is a proposed site for the new cancer research centre. It is located in the same campus as that of the medical college and a cancer research centre.

FEASIBILITY TEST



HOSPITALS IN THE SURROUNDING AREA :

- 2km radius
- 4 km radius
- 6km radius
- 8km radius
- 9km radius

REGULATORY FACTORS

Construction on site follows Kerala Panchayath building rule 2011

- Maximum Permissible coverage(% of plot area)- 40%
- Maximum permissible FAR -3 (Additional Charges)
- The maximum height of the building or part thereof shall not exceed twice the width of
- The street abutting the plot plus twice the width of the yard from the building to the abutting street.

ACCESS FOR OCCUPANCY:

- 1500-6000sqm = 5m
- 6000-12000sqm = 6m
- 12000-18000sqm = 7m
- Above 18000sqm = 10m

PARKING

- 1nos for every 75sqm carpet area.

SETBACK

- (1) The minimum distance between the plot boundary abutting any street other than National Highways, State Highways, District Roads and other roads notified by the municipality and the building, other than a compound wall or fence or outdoor display structure, shall be minimum 1.50 metres.

- (2) Front yard shall have minimum 1.00 metre width.

- (3) In the case of individual developed plots requiring no lay out approval from the District Town Planner, there shall be average 60 cms open space on the sides and the rear:

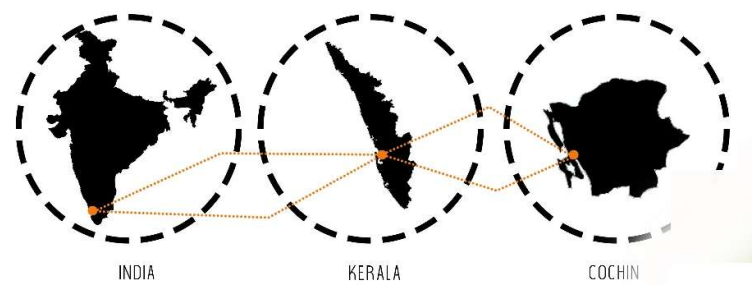
Provided that door shall not be allowed if that portion does not have minimum 1.00 metre open yard.

- (4) In the case of buildings requiring layout approval of District Town Planner, the side yards and rear yard shall be provided as shown in the building layout:

Provided that no portion of the building shall encroach, project or overhang beyond the plot boundaries.

In addition to the minimum front, rear and side open spaces required as per above for height upto 10m proportionate increase in such minimum open space at the rate of 0.50m for every 3.0m height exceeding 10m shall be provided.

SITE ANALYSIS

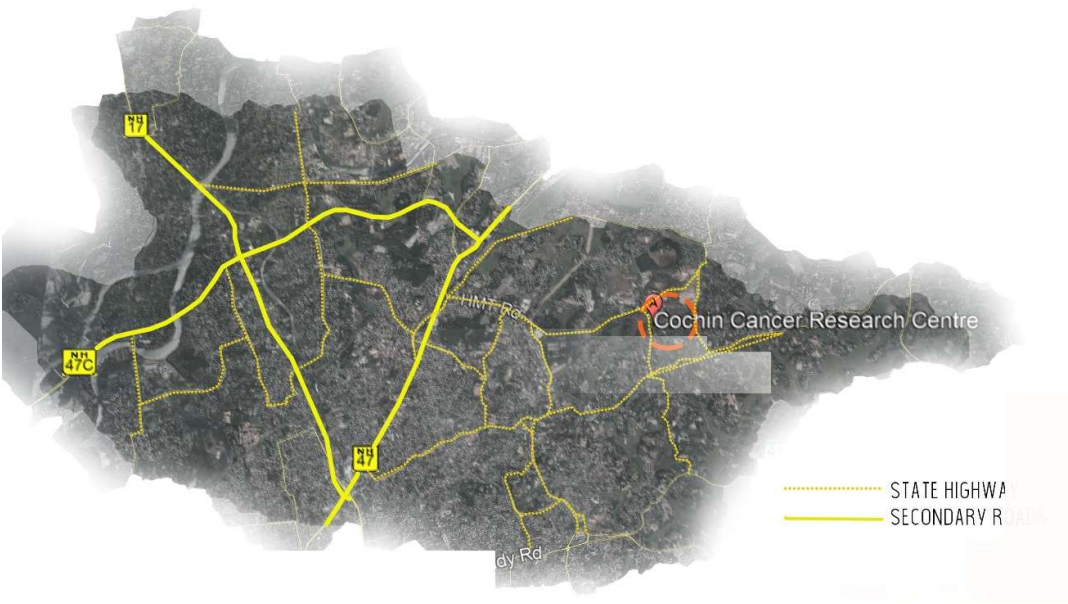


SIZE

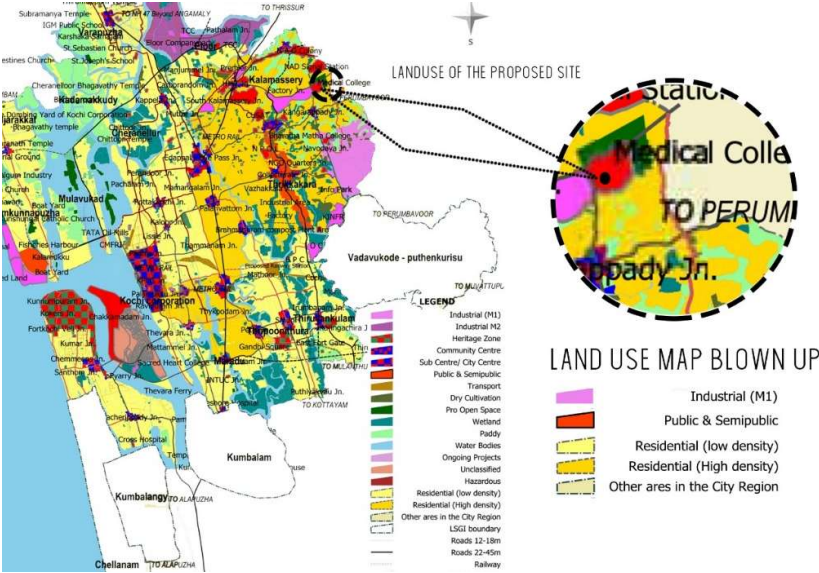


MACRO LEVEL ANALYSIS

LOCATION MAP

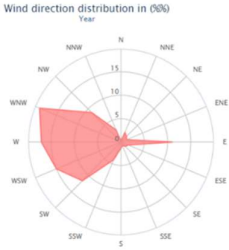
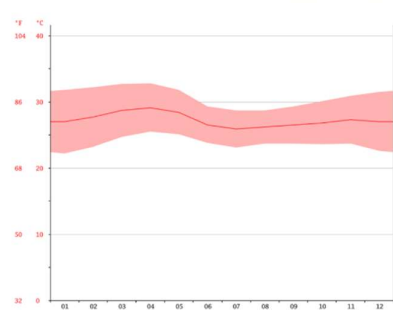
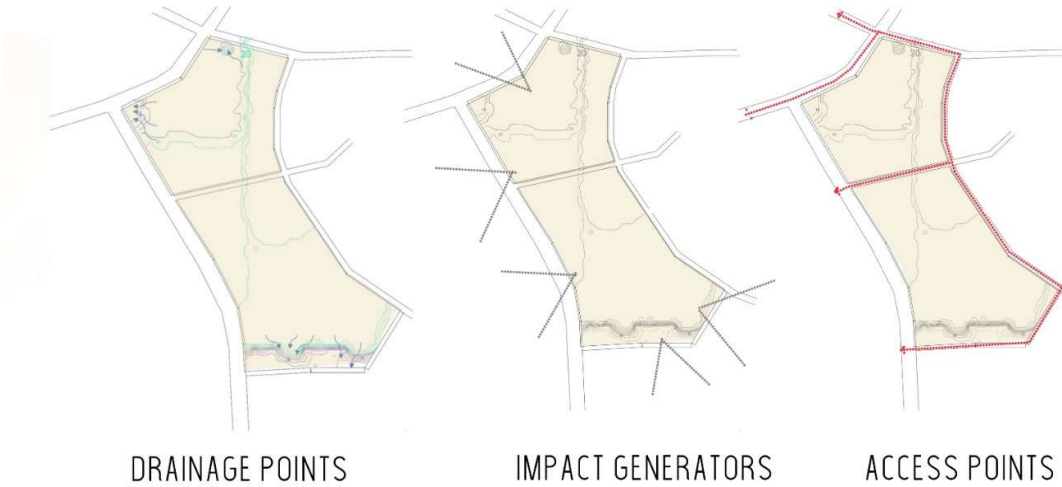
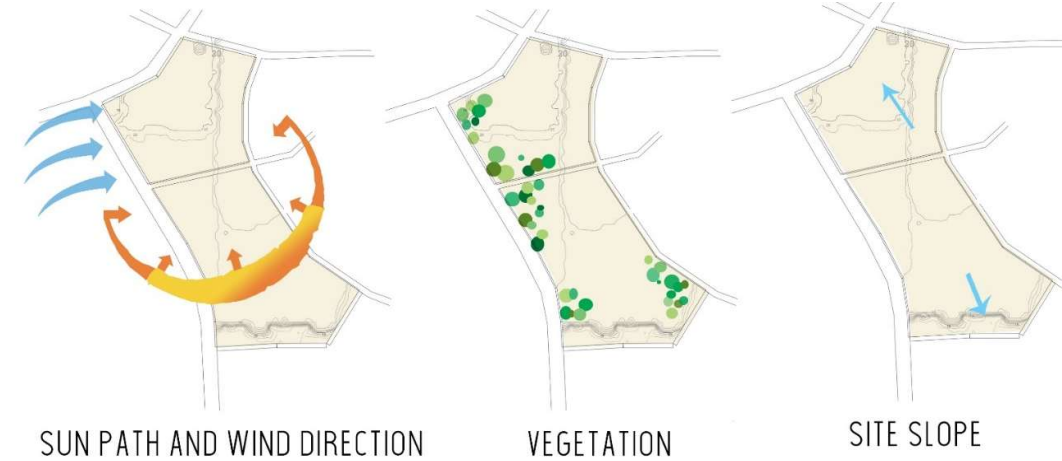


SITE SURROUNDINGS



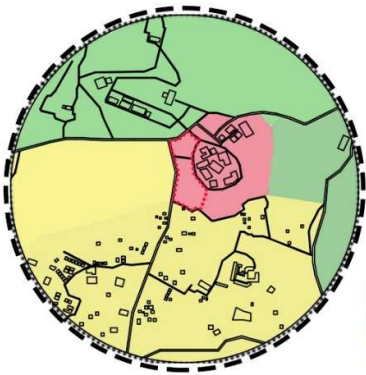
VEHICULAR AND PEDESTRIAN

CLIMATIC ANALYSIS



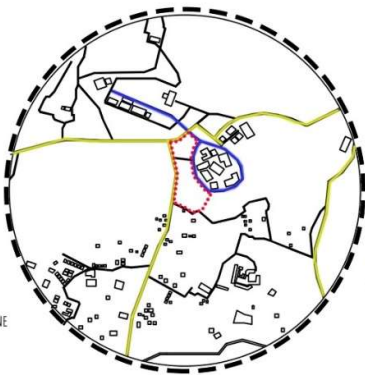
Month of year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
	01	02	03	04	05	06	07	08	09	10	11	12	1-12
Dominant wind direction	➔	➔	➔	➔	➔	➔	➔	➔	➔	➔	➔	➔	➔
Wind probability >= 4 Beaufort (%)	13	19	25	20	18	12	13	16	15	12	8	7	14
Average Wind speed (kts)	8	8	9	8	8	7	7	8	7	7	7	7	7
Average air temp. (°C)	29	30	31	31	30	28	28	28	28	29	29	29	29

MICRO LEVEL ANALYSIS



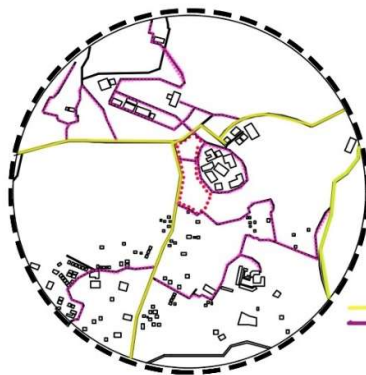
- PRO OPEN SPACE
- PUBLIC/SEMI PUBLIC ZONE
- RESIDENTIAL ZONE

LAND USE MAP



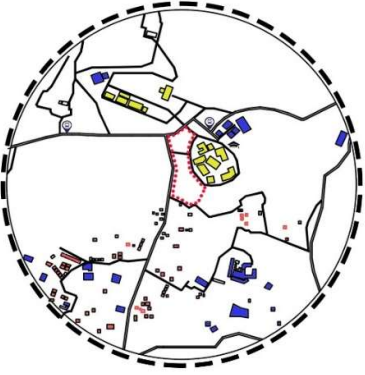
- NATIONAL HIGHWAYS (NH17, NH47)
- MEDICAL COLLEGE ROAD

SITE ACCESS MAP



- ACCESS THROUGH NH
- ACCESS THROUGH SECONDARY ROAD

MOBILITY MAP



- BUILDINGS UPTO 15M
- BUILDINGS UPTO 20M
- BUILDINGS UPTO 10M

BUILDING HEIGHT AND INFRASTRUCTURE MAP

ROOM PROGRAM

ZONES	DEPARTMENTS		SPACES		UNITS	AREA m ²	TOTAL AREA m ²
public	Entry		Trolley park		ST-2 WC-3	9.5	
			Ramp			8	
			General lobby			25	
			General waiting		50	25	
			Toilet		M2+F2+H1	30	
			Gift , book shop			12	
	Reception		Enquiry counter /consultation room			8	
			Queuing tracks		3	16	
			Reg counter			8	
			Record room			25	
			Billing			8	
			Insurance consulting			8	
			E- Registration area		5	12	
	Dispensary		Issue counter		2	8	
			Queuing track		2	9	
			Waiting lobby			20	
			Drug storage			35	
			Staff accommodation			12	
	Emergency unit		Trolley park		ST-2 WC-3	9.5	
			General waiting		25	20	
			Toilet		M2+F2+H1	30	
Semi public			Wards 5 beds			50	
			Causality doctor cabin			16	
			Minor OT			25	
			Blood bank			40	
			Nurse station			10	
public	OPD		Consultation room		10	15	
			Waiting area			25	
			Nurse station			10	
			Consultation area			16	
Semi public			Chemo therapy treatment		10	180	
		IPW	Haematoncology		2	20	
			Paediatric		2	20	
			Day care chemo treatment		15	9	
private			Nuclear treatment	Cyclotron	1	20	
			Anaesthesia		1	20	
			Psychiatric		1	20	
Public			Toilet		M2+F2+H1	30	
			Reception			10	

Semi public	Therapy / diagnostic	Waiting lobby			50	
public		Consultation room		5	9	
private		Changing room		1	7	
		Therapy room	X- ray	1	30	
			CT scan	2	30	
			MRI	2	30	
			Ultra sound	2	25	
			Endoscopy	1	30	
			Mammography	2	30	
			Lenoir accelerator	2	30	
			Recovery room		1	20
		Simulator		1	30	
Dark room		1	25			
Store room		1	45			
Public		Toilet		M2+F2+H1	30	
Public/ semi public	Cafeteria	Kitchen + washing + seating storage	1	500		
Semi public	Wards	Waiting lobby			80	
Private		General ward		2		
		Twin sharing				
		Private ward				
		VIP suits				
		Nurse station				
Semi public		Doctors room				
		Library			50	
Public		Toilet		M2+F2+H1	30	
Public		Dormitory	Waiting lobby			50
	General		30	5		
	Toilet		M2+F2+H1	30		
Private	OT	Changing room				
		Clean corridor				
		Anaesthesia			30	
		Scrub room			20	
		OT				
		Minor OT				
		Toilet				
		Nurse station				
		Doctors buffer room				
		Sterile room				
		Disposal				
		Store room				
		Library				
		Service room				

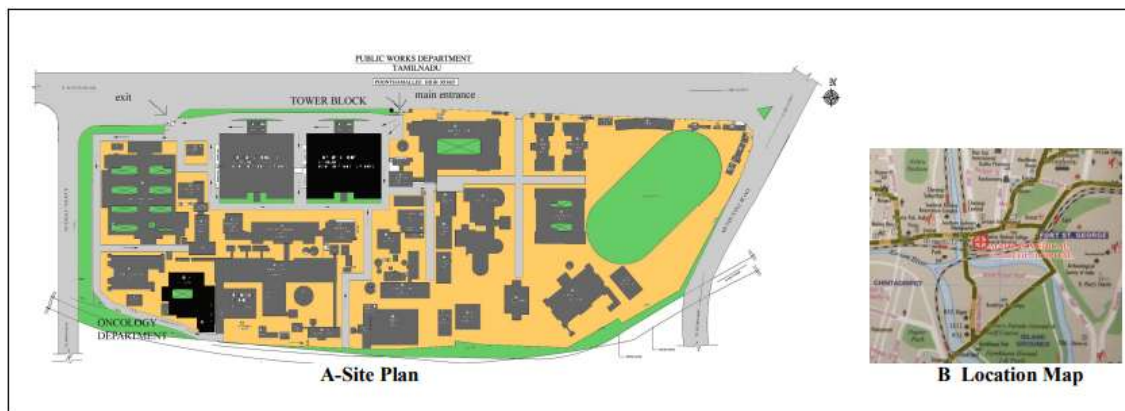
Semi private Private	ICU		Recovery room				
			Post recovery room				
			Surgical ICU				
			High dependent ICU				
			Nurse station				
			Service room				
	Service		House keeping				
			AHU				
				UPS			
				Server			
			Tele oncology				
			Electrical panel board				
			Water treatment				
			Pump room				
			Cylinder room				
			Laundry				
			Locker room				
			Morgue				
		Disposal waste					

CASE STUDY

1. MADRAS MEDICAL COLLEGE HOSPITAL, CHENNAI

LOCATION:

This Institute is a referral hospital with 95% of the cases being referred from various governments secondary and tertiary care government hospitals, namely, the District Hospital and General hospital located in and around the state of Tamil Nadu. (Shodganga)



Source: Public Work Department, Chennai, Tamil Nadu ,2002

SITE PLAN

CASE STUDY 1

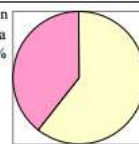
MADRAS MEDICAL COLLEGE HOSPITAL,
CHENNAI

Site area =1,06,580 sq.m

Built-up area=76,490 sq.m

Open area= 50090 sq.m

open
area
40%



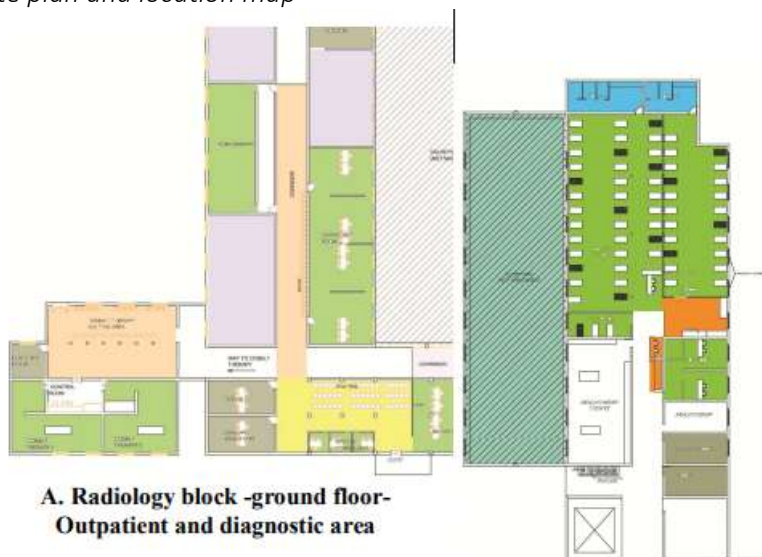
built up
area
60%

Built-up area vs Open area

Drawing not to scale



Site plan and location map





B. Tower block - fourth floor ward

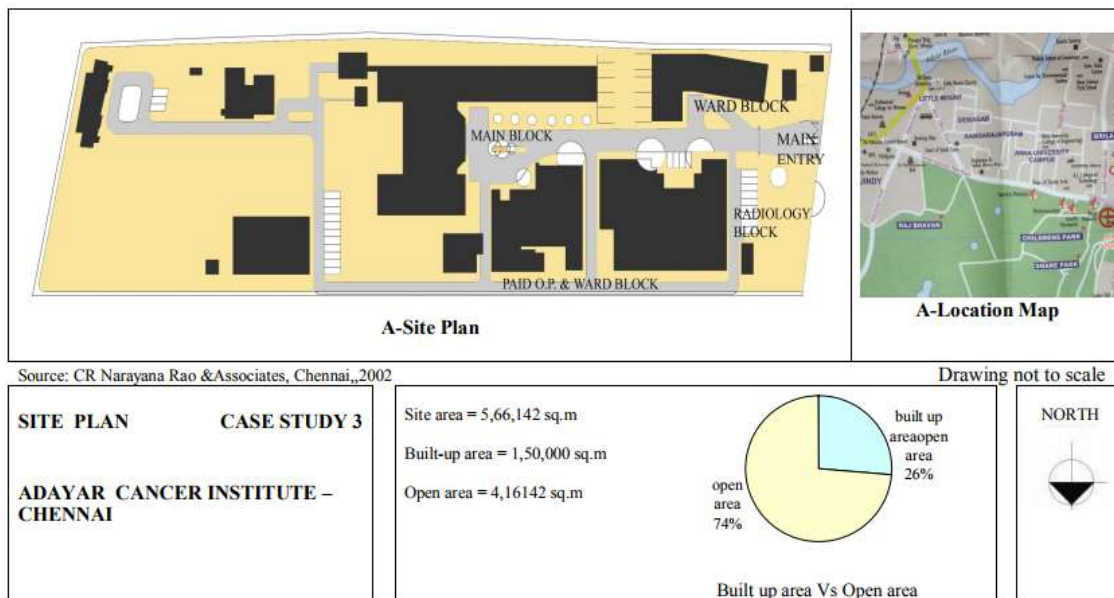
FUNCTIONAL ZONING:

The plan layout of the institute is horizontal spread type with two-storied and three-storied wards block, is located in 13,53,500 sq.m site with the building occupying 7% of site area (Figure A, B). The institute is a horizontal structure with a central corridor spanning across the building from the main entrance, passing through various facilities and terminating in the wards, while the departments are anchored independently on the circuitous corridor with adequate open space between them. The building with less percentage of site coverage and connecting department physical profile provides the indoor-outdoor coalition ideal. The functional hierarchy of the outpatient area and diagnostic area are provided in the front while the patient wards with a singular modular plan are located at the rear with a distinct distance between the male and female wards (Figure A). The operation theatre and administrative sections are located on the first floor followed by additional male and female wards (Figure B) for post-operative care.

2. ADAYAR CANCER INSTITUTE-CHENNAI

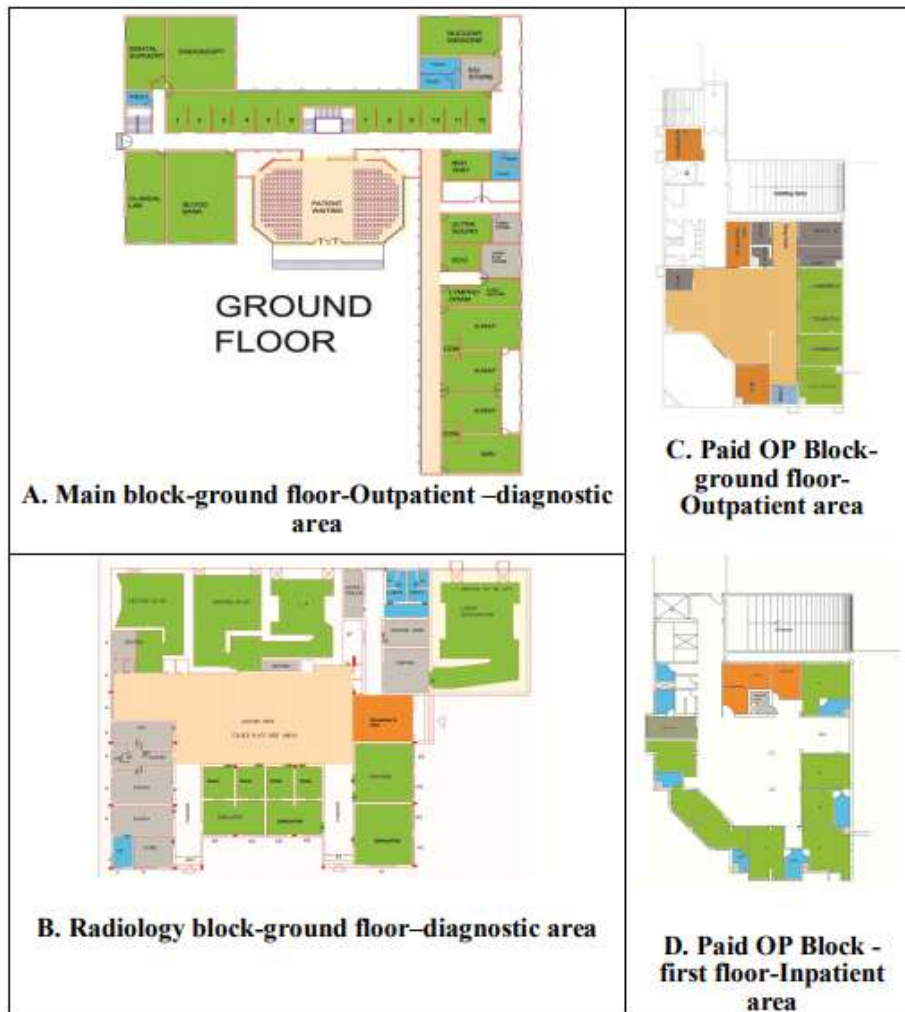
LOCATION:

The Adayar Cancer Institute of Chennai is South India's first comprehensive cancer care centre and the country's second. The Institute has consistently laid emphasis on advanced medical care and research in the field, bringing refined scientific technology and patient care to the poorest in the land. The Adayar Cancer Institute was started in 1954 initially as a small institute in Adayar, a locality in South Chennai; the centre consists of consultation rooms, wards, treatment and surgical units but the linear narrow site limited future developments for more wards and diagnostic facilities. Hence subsequently, the institute moved and developed to a full-fledged centre in the present site at Guindy situated in the institution zone of the city (Figure A, B). Today, the whole functioning in the old site has been shifted to the new site, with the former only housing the free wards with minimal staff.



FUNCTIONAL ZONING:

The plan layout of the institute is horizontal spread type with two-storied and three-storied wards block, is located in 13,53,500 sq.m site with the building occupying 7% of site area (Figure A, B). The institute is a horizontal structure with a central corridor spanning across the building from the main entrance, passing through various facilities and terminating in the wards, while the departments are



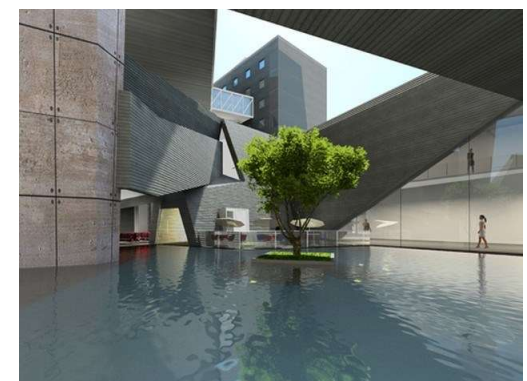
anchored independently on the circuitous corridor with adequate open space between them. The building with less percentage of site coverage and connecting department physical profile provides the indoor-outdoor coalition ideal. The functional hierarchy of the outpatient area and diagnostic area are provided in the front while the patient wards with a singular modular plan are located at the rear with a distinct distance between the male and female wards (Figure A). The operation theatre and administrative sections are located on the first floor followed by additional male and female wards (Figure B) for post-operative care.

3. VEDANTA CANCER CENTRE- RAIPUR

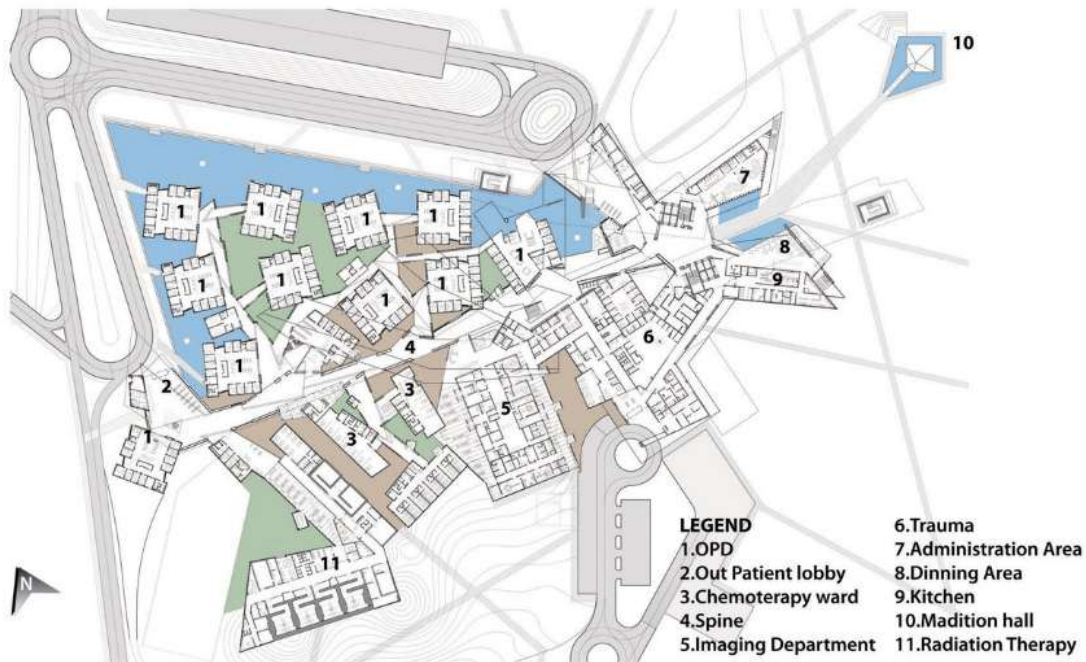
The Vedanta Institute of Oncological Sciences is designed to be, first and foremost, patient -centric. Critical movement paths of patients/doctors/services/visitors and 3-dimensional functional interconnectivity have been synergistically established as an outcome of the workshop. The focus rests on the patient, at both a physiological and psychological level, and although the effect of nature, green spaces and water is not quantifiable, they have been integrated and emphasized at every level of planning. This facility represents a convergence and merging of science and spirituality, and nowhere is this more symbolized than at the extremities of the building spine. At the northern end, a subterranean meditation centre is accessed through trees and a water-body and captures the first rays of the sun. At the western end is a teaching facility, library and auditorium. It is a representation of our collective acknowledgement that, when confronted with the difficult and ambiguous issue of our own impermanence, the journey within can often provide answers that external sources often fail to yield.

An east-west spine has been developed as a street flanked by all the out-patient and diagnostic facilities, and containing all common-visitor facilities (ATMs, pharmacy, phone booths, restrooms, cafes etc) along its length. The main out-patient entrance and registration is designed to process and disseminate a large number of referrals, and to surmount the inevitable problem of communication and navigation that occurs in country with multiple languages and hundreds of dialects, colour coded cards are handed out so patients can follow graphic vectors on the floor to their destination.

Individual out-patient pods corresponding to different parts of the human anatomy (this was one of the points that all specialists concurred on) are arrayed in a north-facing water-body, interspersed with landscaped courts and connected to the street via sub-spines. These pods are differentiated by their modified roof profiles, manipulated individually within a common structural framework to bring north light into the waiting areas, a principle that extends into the main street and the sub-spines. The areas in the north are oriented to make the most of the soft natural light, and a journey to the pods inevitably leads you past landscaped break-out areas.



GROUND FLOOR PLAN



Major diagnostics, radiotherapy and the proposed proton accelerator are functions that require very restricted natural light and are therefore arrayed on the south of the spine as solid bunkers, where they shield the spine and the intermediate courtyards from the harsh south sun. The chemotherapy and imaging blocks are placed further east along the spine and fragmented courts bring natural light into working and waiting areas.

The street ensures highly streamlined patient movement whilst always keeping them in contact with nature. A service tube runs below the spine to carry services through the length of the facility and branches out at critical locations to provide tap-offs.

The inpatient areas are accessed by a separate entry in the north-east, where a large lobby with common facilities acts as a node to channel restricted movement further into the facility. Certain facilities, like a lounge with slumberettes, were added to contextually adapt the in-patient areas to Indian sociology. The in-patient wards are stacked to optimize services and circulation.

The 1st level of the facility is restricted primarily to the doctor and staff areas, and these have been translated as diagrams connecting volumes to critical points of contact below. The operation theatres, surgical ICU's, and medical ICU's are also located at this level, as is a compact cardiac facility. (Super-specialty cancer centres are considered redundant as cancer affects all parts of the body, and in



order to be self-sustaining, this facility needed to be equipped to handle all possible contingencies. Other parts of the anatomy could be dealt with, but cardiac complications required some additional attention).

A thorough study was done to ensure the sustainability of this building. All flat roofs are being developed as green areas (this also ensures that even areas like operation theatres look out onto green spaces), solar panels have been used extensively on south and west facing areas. Orientations have been studied for glazing and wind movements. The major circulation areas have been designed for natural ventilation in good weather, and free-cooling in warm weather (temperatures of up to 45 degrees can be reached.) Rain water harvesting, geo-thermal energy and a host of other measures have been taken.

4. WINDER REGIONAL CANCER CENTRE, ONTARIO

Located at the southwest edge of the Windsor Regional Hospital's Metropolitan campus, the building is sited to address the neighbouring residential streets and tree lined boulevards of Kildare Road and Alsace Avenue. The centre is based on a three radiation treatment room module of approximately 6500m².

Basic Organization the building is spatially organized over three storeys. The main entrance is located on the south side of the building off of Alsace Avenue. Patient parking is provided adjacent to this entry. A pedestrian entrance from Kildare Road is provided on the west side for visitors using public transit. Ambulances arrive at the hospital's ambulance bay located east of the centre.

The building's section follows the existing grade to the south at the entrance at Alsace and then drops approximately one meter, to align with the Windsor Regional Hospital first floor level. The change in floor level creates a natural separation of public and clinical spaces. The building's sectional design was developed to spatially interconnect the patient treatment floors, levels one and two.

A ramp along the western edge of the lobby leads patients and visitors down to the radiation treatment area. An open staircase leads patients up to the outpatient clinics and the chemotherapy suite overlooking the entry garden. Once past the main lobby, patient circulation follows the western edge of the building, along which are located a series of waiting and counselling areas with extensive day lighting and garden views. Along this route are the departments of Patient Review and Simulator. At the extreme north end are the radiation suites. The suites are accessed through a two storey high corridor with north facing clerestory glazing.

Design Inspiration and Approach The project reflects the Client's particular concern for patient focused care and the design team's interest in reinforcing the site context. Overall the design strives to create a state of the art cancer facility that instils confidence in the care being delivered and to provide an environment that is comforting and promotes a sense of life, hope and harmony with nature. Providing a regional identity, integrating the building into the surrounding residential neighbourhood and reinforcement of the existing mature landscape were important influences in the design decisions.

LANDSCAPE:

The designers were fortunate to have several large street trees on the property. These trees help to mitigate the scale of the building, provide shade and make the new landscape areas appear more mature. New gardens were created along the building's south entry façade and west façade, adjacent to patient circulation. The gardens not only provide interesting views, they also greatly assist patients in way finding.

A partially walled entry garden was constructed on the southern side of the building. The garden features a linear water garden that serves as a calming space to visit or pass by upon entry of the facility. The garden is viewed from the main lobby, the waiting area for specimen collection and from the chemotherapy suite on the second floor.

A sunken garden along the west façade follows the patient circulation routes to the treatment areas. As Windsor has the longest growing season in the province of Ontario, the gardens contain a wide variety of plant specimens including several healing plants used in cancer treatments.

The gardens were designed by the architect, landscape architect, healing plant designer, plant designer, water garden specialist and lighting designer. Businesses and the local community provided partial funding for the gardens.



INTERIORS:

The interior design builds on the notion to incorporate the essential elements of life into the public areas of the building:

AIR, WATER, FIRE AND EARTH.

- **WATER:** The entry sequence brings visitors along the entry garden where the element of water is the overwhelming focus.
- **FIRE:** Once inside the main entry lobby, the focus shifts to a freestanding brick and stone fireplace.
- **AIR:** A feeling of loftiness is created in the public areas. They are a minimum of five meters in height and have clerestory glazing in two orientations.
- **EARTH:** All public areas and major circulation routes follow the perimeter of the gardens.

As the ground floor level is one meter below grade, the sunken garden accentuates the sense of shelter and being part of the earth. The interior design strives to balance the highly technical nature of the treatment equipment with the basic elements inherent in natural materials and

daylight. Clay brick, limestone and Douglas fir, used on the building exterior, are carried through to the interior. Paint and plaster colours of ochre and green were selected to enhance the connection with the outdoor landscape and contrast with an extensive use of rich cherry panelling. Careful consideration has been paid to the design of the ceiling plane and light quality. The designers have used fluorescent light sources in indirect applications only. Accent lighting is provided at feature walls and low level lighting provided in patient corridors and at interior brick piers.

THE DESIGN PROCESS

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